

X3 ATOMIC STAR THE ROAD TO MAINNET

Evidence-Based Architecture Audit, Gap Analysis, and Production Completion Blueprint

AUDITED EVIDENCE READINESS

20 / 100

PUBLIC TESTNET: NO-GO

MAINNET: NO-GO

Three Critical findings remain open.

05 SEPTEMBER 2026 / MASTER + EXISTING EDITS

Codex / AI-assisted analysis

Base commit: 6a24d8cf38f2522ddf9ae0b47011fd59a9984208

Read-only protocol audit • Evidence package • Completion blueprint

Scope, provenance and how to read

Technical engineering review • AI-assisted • Not independent security certification

Intended audience: protocol engineers, security reviewers, validators, testnet operators, grant/sponsor reviewers and future contributors. The base commit is 6a24d8cf38f2522ddf9ae0b47011fd59a9984208 on master, with extensive pre-existing tracked and untracked edits. The audited source is not a clean checkout of that commit.

Safety and scope: protocol code was inspected without modification. Safe local builds/tests used an isolated source copy, offline Cargo and disposable outputs. No deployments, public-network signing, transactions, live-fund interaction or production credentials were used. Audit artifacts are the only intended repository changes.

The review gives an evidence-based readiness decision and completion plan. It is not independent security certification, a formal cryptographic proof, a line-by-line review of every vendored or first-party file, or a claim that blocked experiments passed. Execution, static inspection, inference and documentation claims are distinguished.

Reading order: start with Chapter 1, then Chapter 5 for actionable defects. Engineers should use Chapters 7-12 and the finding/feature JSON/CSV files. Operators should use Chapters 11 and 14. Sponsors can use executive-summary.md and Chapter 13. Every numbered figure is a reusable vector asset; all measurements and scores derive from shipped data.

Evidence references such as C01, H18, FT01 and file.rs:123 lead to the finding register, feature scorecard and exact audited source. PDF page numbers include front matter. Checksums and final page count are recorded externally after rendering, avoiding circular self-hashes.

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DECISION / EVIDENCE / PRIORITIES

1 Executive Command Brief

X3 Atomic Star is a substantial Substrate-based protocol codebase with Aura/GRANDPA node wiring, a FRAME runtime, internal asset representations, atomic orchestration, external proof routes, language tooling and multiple applications. The inspected state is an internal development candidate. It is not ready for an open public testnet or a mainnet handling value.

FIELD MANUAL / 01

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

PUBLIC TESTNET: NO-GO | MAINNET: NO-GO | READINESS: 20/100. Three Critical findings remain open. This evidence score is not a percentage of code written or a probability of safety.

If you read nothing else

The most serious problem is misplaced trust at protocol boundaries: header, finality-anchor and rollback-receipt paths accept caller assertions where authenticated evidence is required. Runtime cross-chain header validation ignores the expected root (C01); unsigned nonzero hashes can become finality anchors (C02); unsigned callers can supply the prior state used during rollback (C03). These findings are grounded in reachable runtime wiring, not merely dormant example code.

An isolated test harness compiled the unchanged proof-router source with its production feature. All three rejection tests failed because unverified payloads were accepted (H01). A separate four-test RPC algorithm harness passed. Selected Python DSL tests returned 6 passed and 4 failed. Whole-workspace check/test/clippy and the release/testnet feature builds all failed in the WASM build with E0152 duplicate core. No live transaction, finalized network, contract deployment or restore was demonstrated in this audit.

The register contains 29 findings: 3 Critical, 18 High, 7 Medium and 1 Low. Confidence is high in the NO-GO decision, high in directly observed helper/gate failures, and moderate in broader system completeness because full node execution was blocked. Findings are not a claim that every file has been manually reviewed or every possible vulnerability discovered.

Table 1 Five strongest foundations

Foundation	Evidence and exact limit
Real standard consensus wiring	node/src/service.rs:534, :1059, :1136: Aura import/authoring and GRANDPA voting are present; live safety/liveness not verified.
Native transaction controls	runtime/src/lib.rs:850: nonce, genesis, era, weight and transaction-payment extensions are explicitly wired.
Accounting and recovery intent	Supply-ledger checked arithmetic and economic-halt paths exist; C03/H16 and multi-node verification remain open.
Executable language and RPC units	Parser/typechecker/CLI and four middleware unit tests executed; this does not prove cross-chain swaps.
Useful operational assets	Pinned Rust toolchain, systemd hardening, explicit scopes and lockfiles provide a basis for reproducible delivery; current gates/build fail.

Table 2 Five most dangerous weaknesses

Risk cluster	Consequence
C01 + H01/H02/H03	Unauthenticated external truth can be treated as proof.
C02 + C03	Finality and rollback claims have untrusted inputs.
H04/H05/H06	Validator penalties, rotation and experimental finality are incomplete or unsafe.
H07/H08/H09/H11/H12	Execution, submission, gateway and SDK paths do not consistently deliver the advertised protocol.
H13/H14/H15/H18	Build/gate/advisory evidence can be missing, stale, suppressed or misleading.

Figure 24 shows overall readiness evidence.

Overall readiness evidence

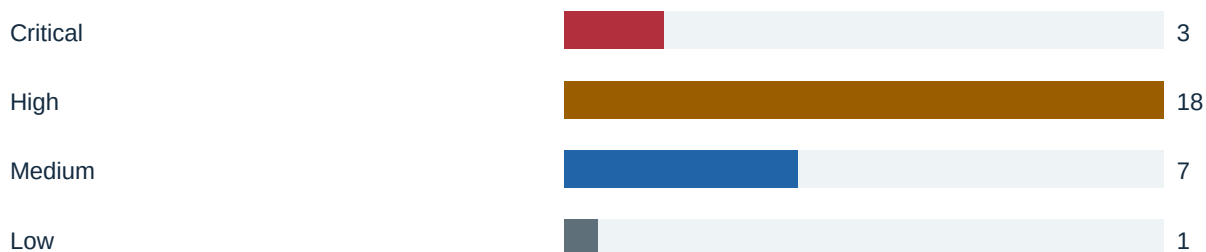


Any unresolved Critical finding caps readiness at 20; this is not a probability of safety.

Figure 24 Overall readiness evidence

Figure 3 shows finding severity distribution.

Finding severity distribution



Review findings, not a count of independent exploits.

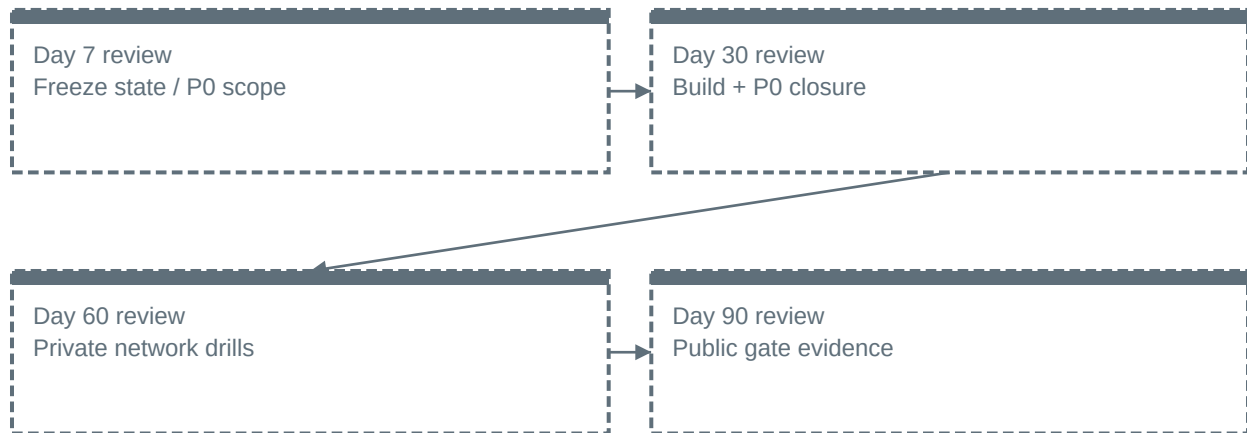
Figure 3 Finding severity distribution

Fastest credible route forward

Freeze this source state, repair the build without skipping WASM, fix failure propagation, and close C01-C03 before spending effort on throughput or public launch. Demonstrate one signed native transfer through finality and restart. Then add one internal atomic route with conservation and rollback proof. External chains should be separate gated milestones after authenticated proof implementations and independent review. No public-network signing or deployment was performed.

Figure 13 shows 7 / 30 / 60 / 90-day planning horizons.

7 / 30 / 60 / 90-day planning horizons



I implemented/unverified P partial D disconnected X unsafe V verified N proposed

Review horizons only; dependent on staffing and acceptance gates, not delivery promises.

Figure 13 7 / 30 / 60 / 90-day planning horizons

The 7/30/60/90 labels are review horizons, not promises. Planning assumes access to at least protocol, security, build/release and distributed-test expertise; no team size or committed allocation was supplied. Re-estimate after the first build and adversarial closure milestone. Feature expansion must not outrun proof quality.

CODE PATHS / TRUST BOUNDARIES

2 System Architecture

Architecture is reconstructed from Cargo metadata, binary entrypoints and inspected runtime configuration. Solid boxes represent code/wiring; they do not mean successful execution. Dashed boxes mark disconnected or proposed paths. All diagrams are reusable SVG/PDF assets in assets/.

FIELD MANUAL / 02

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
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Figure 7 shows system architecture.

System architecture

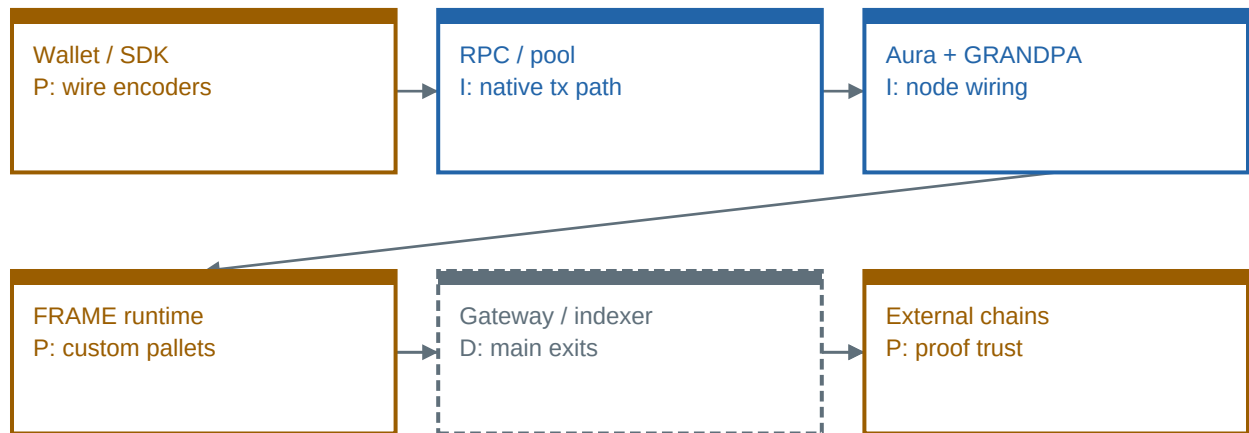


Figure 7 System architecture

Figure 11 shows component dependencies.

Component dependencies

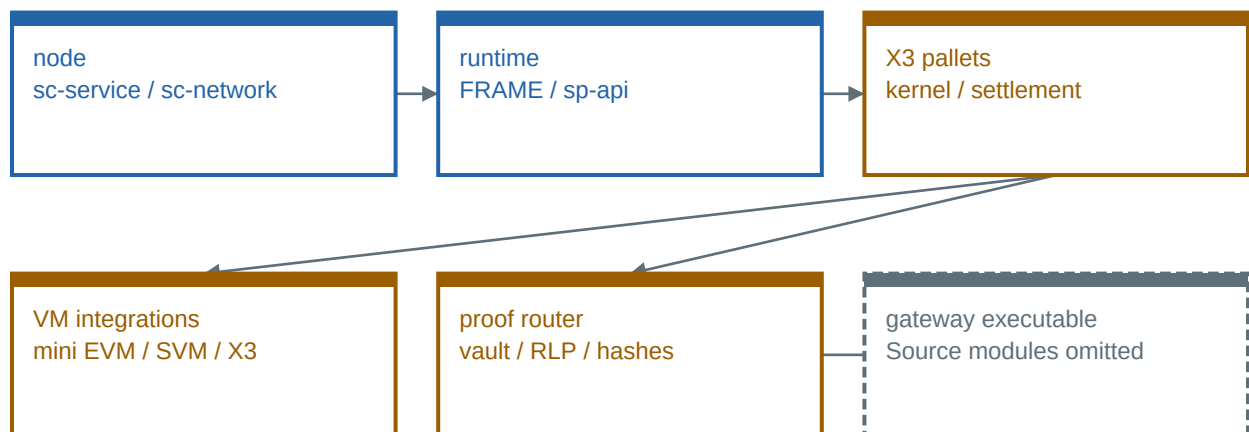
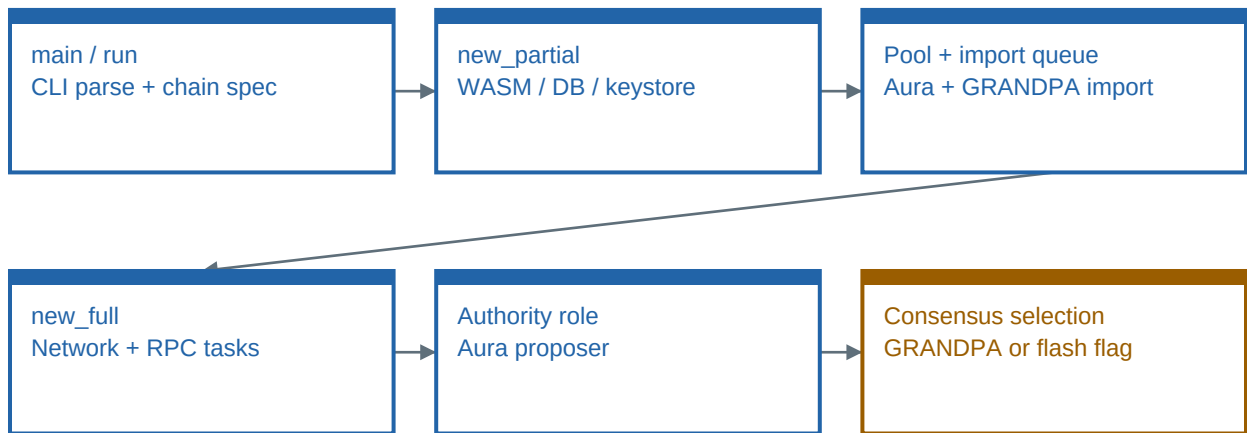


Figure 11 Component dependencies

Figure 14 shows node startup path.

Node startup path



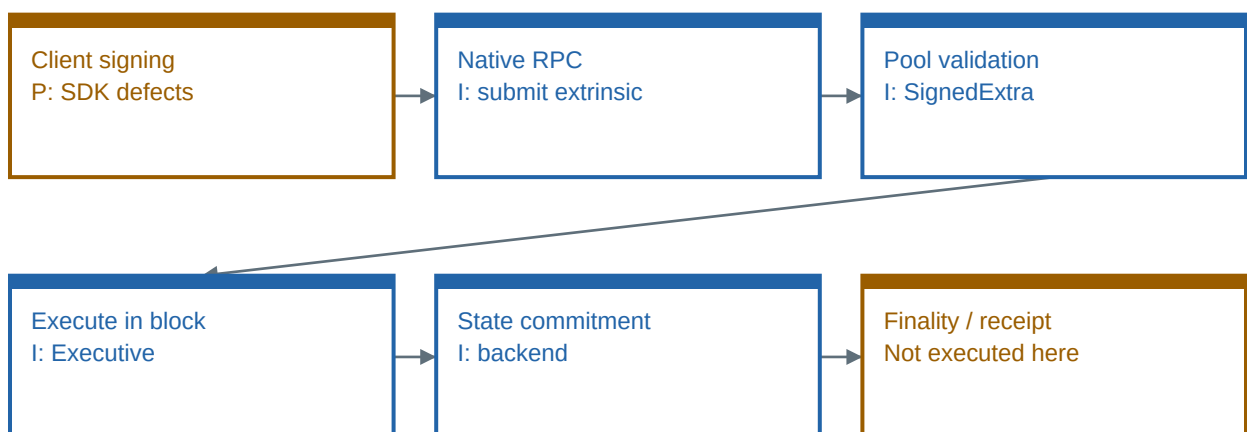
I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 14 Node startup path

Startup begins at `node/src/main.rs:4`, delegates to `x3_chain_node::run`, parses CLI in `command.rs`, selects/generates a chain spec and invokes service factories. `new_partial` constructs the WASM executor, backend, keystore, pool, longest-chain selector, GRANDPA block import and Aura queue. `new_full` builds network/RPC tasks and starts Aura for authority roles. Default finality is GRANDPA; experimental Flash opt-in changes that choice (H06).

Figure 8 shows transaction lifecycle.

Transaction lifecycle



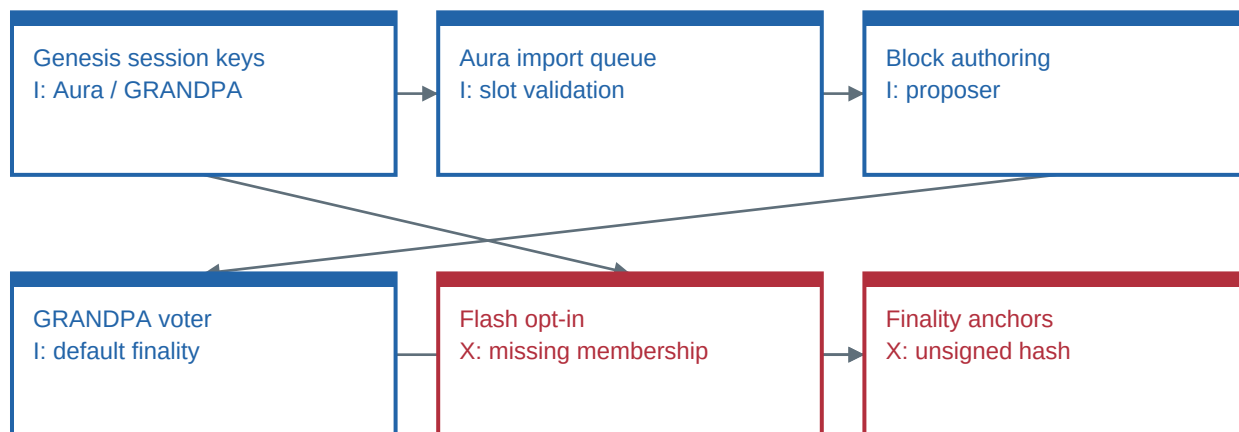
I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 8 Transaction lifecycle

The ordinary native path is a signed SCALE extrinsic submitted through author RPC to the transaction pool, validated through runtime APIs and SignedExtra, executed by Executive during block production/import, committed into FRAME storage and then finalized by GRANDPA. Receipt/event/indexer consumers are a separate path. This is a static trace: the full transaction lifecycle could not be executed because the current runtime build fails. The Ethereum-named submission method deviates from this lifecycle (H08).

Figure 9 shows consensus and finality.

Consensus and finality

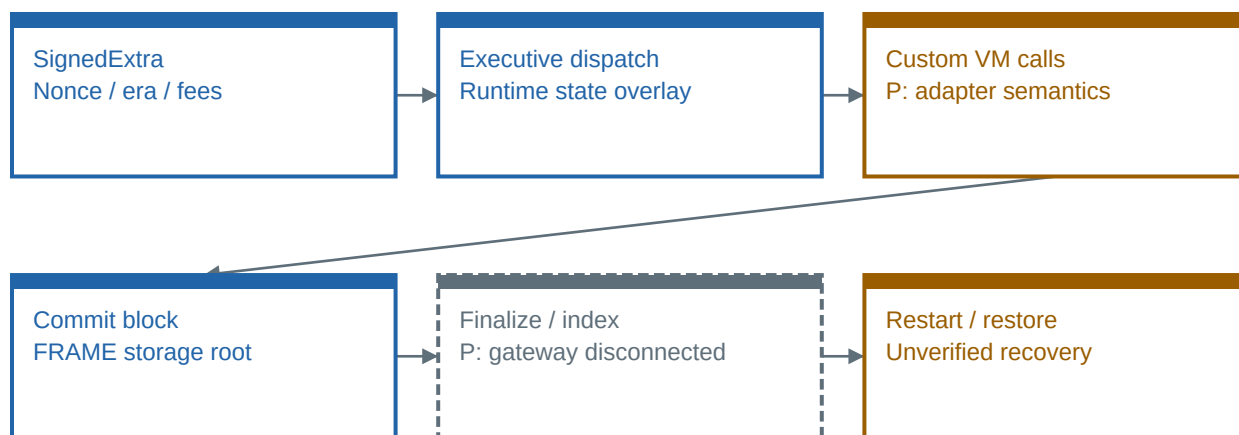


I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 9 Consensus and finality

Figure 15 shows state transition and persistence.

State transition and persistence

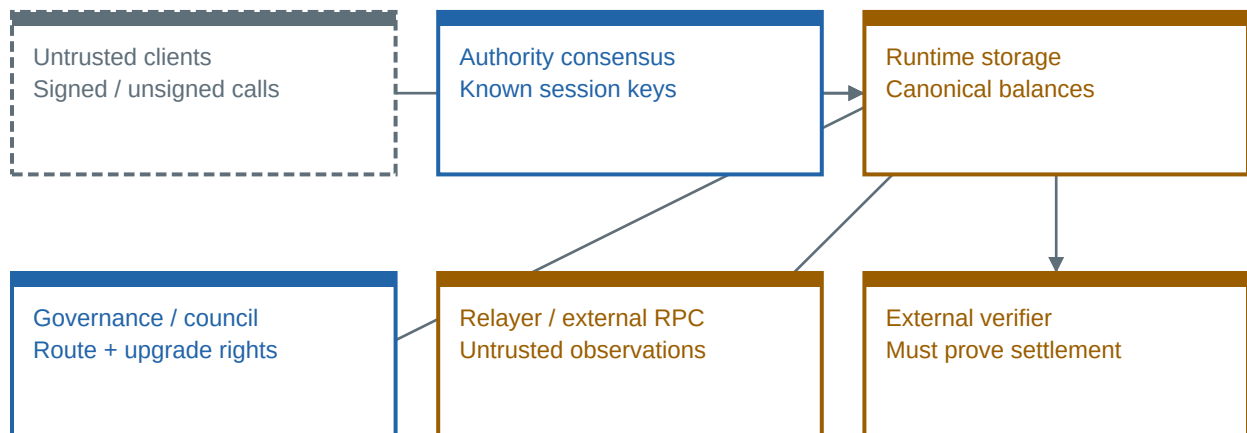


I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 15 State transition and persistence

Figure 10 shows trust boundaries.

Trust boundaries

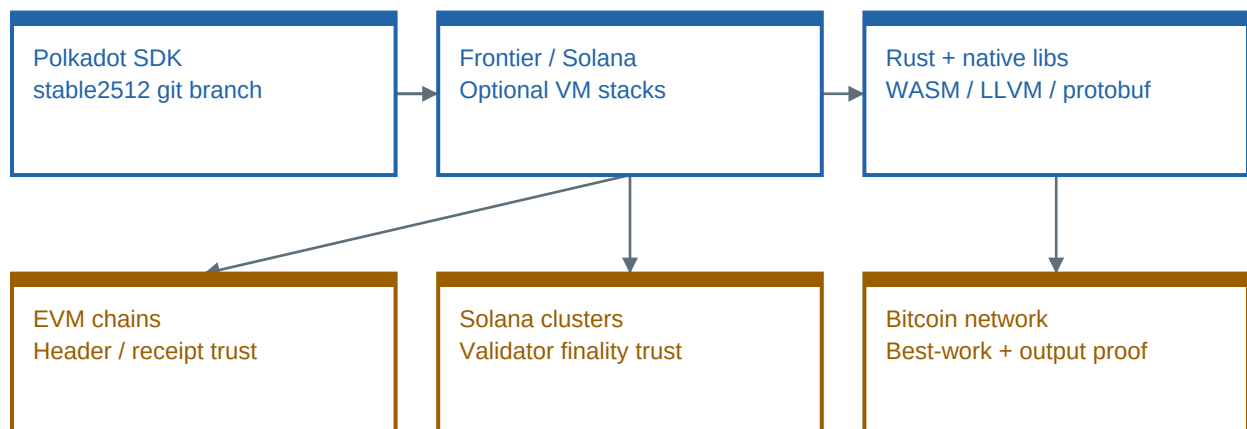


I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 10 Trust boundaries

Figure 17 shows external dependency map.

External dependency map



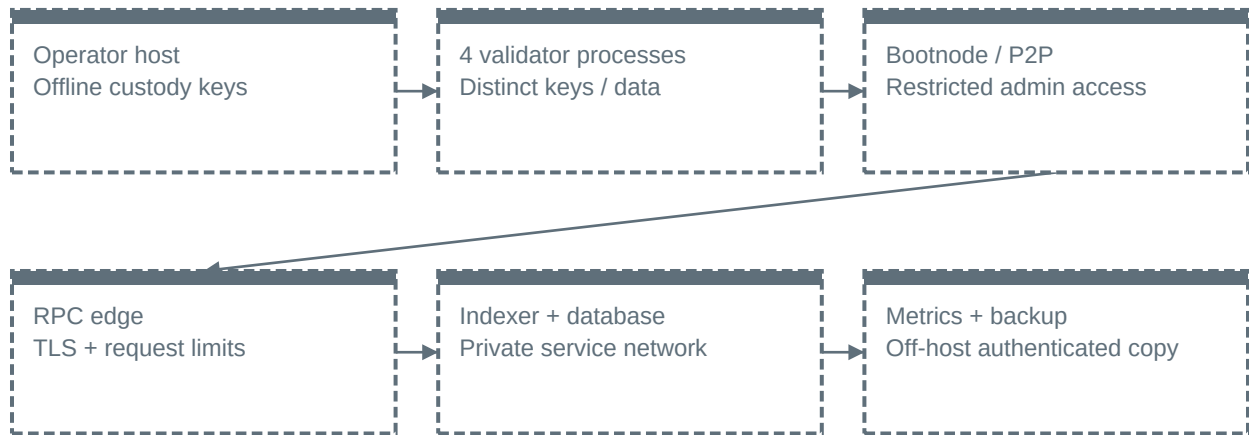
I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 17 External dependency map

External trust includes Polkadot SDK/Frontier/Solana dependencies, configured session authorities, governance/council decisions, relay/custody services, RPC observations and contract verifier owners. A root/half-council origin currently guards gateway proof operations (runtime/src/lib.rs:2479); this limits direct unprivileged access but is not a cryptographic substitute for verification. The older cross-chain header pallet still exposes signed submission, and atomic anchors/diffs expose unsigned calls.

Figure 16 shows proposed isolated testnet topology.

Proposed isolated testnet topology



I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Proposed topology for validation; no live infrastructure was contacted or deployed.

Figure 16 Proposed isolated testnet topology

INVENTORY / BUILD GRAPH / REACHABILITY

3 Repository Anatomy

The first-party inventory contains 11,620 paths, 375 manifests and 145 Cargo workspace packages. Cargo.lock resolves 1,996 package records. Inventory coverage and manual review are distinct. Vendored dependency source, legacy snapshots, credential files and build/install outputs were not treated as first-party implementation evidence.

FIELD MANUAL / 03

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Table 3 Repository groups and responsibilities

Path/group	Responsibility / evidence
node/	x3-chain-node executable: CLI, chain specs, service, network and RPC.
runtime/	x3-chain-runtime WASM, FRAME pallet composition, signed extensions, runtime APIs and migrations.
pallets/	Consensus metadata, balances/asset representations, kernel, settlement, governance, wallet, DEX, supply and bridge state machines.
crates/	Off-chain services and libraries: VM/compiler, proof routes, relayer, gateway, attestation, finality and health tooling.
x3-lang/	Separate Python DSL and Rust language workspace; source-of-truth claims differ from experimental root compiler track.
X3-contracts/; programs/svm/	Separate Foundry/Anchor workspaces and additional SVM programs. They are not covered by a root workspace test alone.
apps/; packages/	Desktop/Tauri, web frontends, wallets, SDKs and transport adapters; root npm scripts chain many package commands.
services/; swarm/; infra*/	Sidecars, Python services, infrastructure and database consumers. Deployment examples are not evidence of deployed services.
scripts/; proof-forge/; proving/	Launch gates, integrity scanners, proof-generation tools and proving harnesses. Their claims must be compared with executable behavior.

Table 4 Code exists versus code runs

Component	Exists	Runtime reachability
Node core	Yes: service/consensus/backend source	Build blocked in this audit; no node launch claimed.
Gateway REST/GraphQL	Yes: rest.rs, db.rs and schema modules	Binary main logs and exits; modules are not declared/served (H11).
Proof router	Yes; compiled by isolated harness	Unsafe production alternatives accepted invalid proof bytes (H01).
Legacy EVM structural verifier	Yes, compiled with production feature	Not the selected production EVM gateway route; exposed library remains unsafe.
Node bootstrap helper	Yes: BootstrapConfig	No demonstrated service consumer of this helper; actual sc_network configuration is separate.
Python parser/typechecker	Yes	CLI and relevant tests executed successfully.
Two-node finalized bridge test	Yes	Ignored; no new network-level proof supplied.

Complete component-map.json records package targets/features/dependencies; inventory.json records standalone manifests/scripts and path categories. These machine-readable inventories cover more components than can be usefully typeset as an architecture diagram. The feature scorecard scopes the behaviors actually assessed, while unreviewed minor code remains unknown rather than silently receiving credit.

Table 5 Migration and configuration inventory

Category	Observed scope / next proof
Runtime migrations	runtime/src/lib.rs:826 wires kernel, treasury, agent-memory and agent-accounts migrations. Other migration.rs files require per-pallet hook/version review; presence alone does not prove they are scheduled.
Gateway SQL	Database query/migration source exists but gateway main is disconnected. Need fresh schema + prior schema upgrade against disposable PostgreSQL.
Python migrations	tests/test_migrations.py imports alembic and swarm.db. Collection failed at missing alembic; no schema operation was run.
Config flow	CLI/sc_service config to node; live authorities/council/treasury/escrows from environment into chain spec; relayer YAML/env fallback in its binary; gateway config currently disconnected.
CI	39 workflow paths inventoried. H13 demonstrates swallowed recipe failure and missing workflow target; hosted required-status settings were not queried.
Deployment	232 first-party deployment-category paths and additional systemd/compose assets inventoried; no cluster or live service was contacted.

TRANSPARENT CRITERIA / NO DOCUMENTATION CREDIT

4 Feature Completion Scorecard

Each feature = $20 \times (\text{implemented} + \text{wired} + \text{tested} + \text{executed} + \text{reproducible})$, indicators 0/1. Tested means a passing applicable test observed in this audit. Scoped algorithm/DSL verification is not whole-network verification. Overall = weighted subsystem mean, capped at 20 while any Critical finding is open.

FIELD MANUAL / 04

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

An indicator gets 1 only for the stated feature scope. Implementation means meaningful behavior exists for that scope; wiring means the inspected entrypoint calls it. Passing tests, executed behavior and reproducibility need current audit evidence. Missing/blocked evidence gets 0 without implying code is absent. Two VERIFIED rows have deliberately narrow scopes: RPC limiter algorithms and Python parsing/typechecking. Whole-node RPC protection and complete language execution are separate incomplete rows.

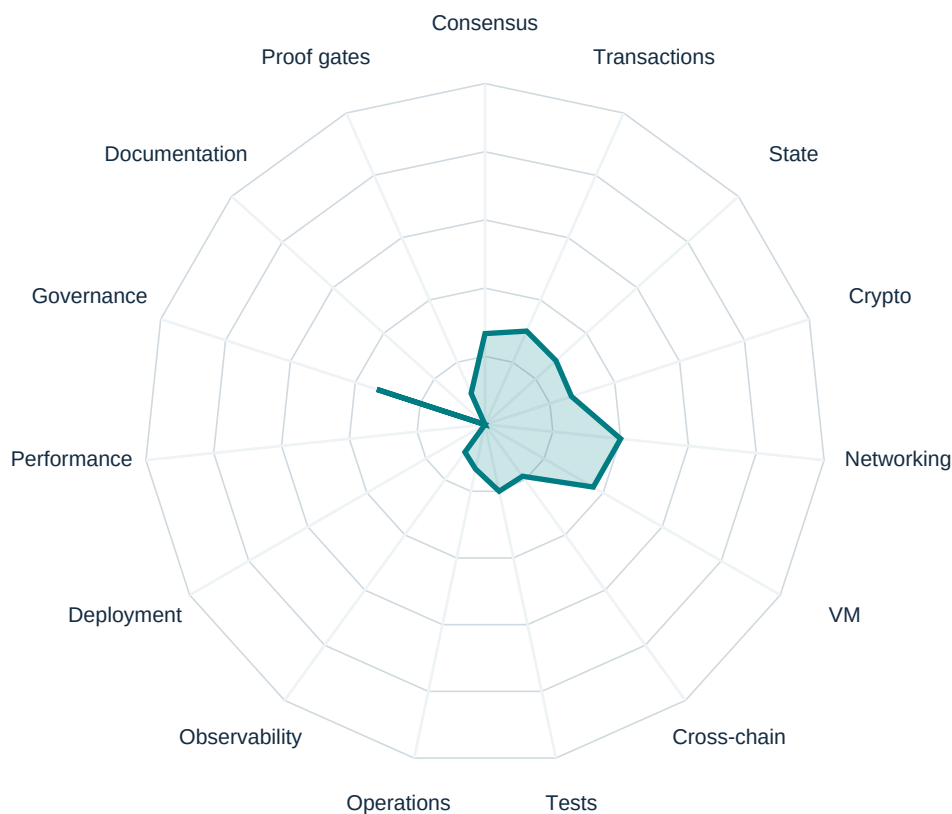
The uncapped weighted mean is 24.29/100. The published readiness score is 20/100 after applying the explicit safety cap of 20 for any open Critical finding. This policy prevents numerous peripheral features from outweighing broken safety boundaries. Weights sum to 100. All feature indicators, weights, references and uncertainty appear in scorecard.json and the full CSV.

Table 6 Subsystem scoring weights and raw evidence

Subsystem	Weight / 100	Raw score / 100
Consensus	14	26.67
Transactions	10	30.0
State	12	28.0
Crypto	10	26.67
Networking	7	40.0
VM	8	36.67
Cross-chain	14	18.75
Tests	5	20.0
Operations	5	13.33
Observability	3	10.0
Deployment	4	0.0
Performance	2	0.0
Governance	3	33.33
Documentation	1	0.0
Proof gates	2	10.0

Figure 5 shows readiness radar: equal axes, evidence-derived raw scores.

Readiness radar — evidence, not probability

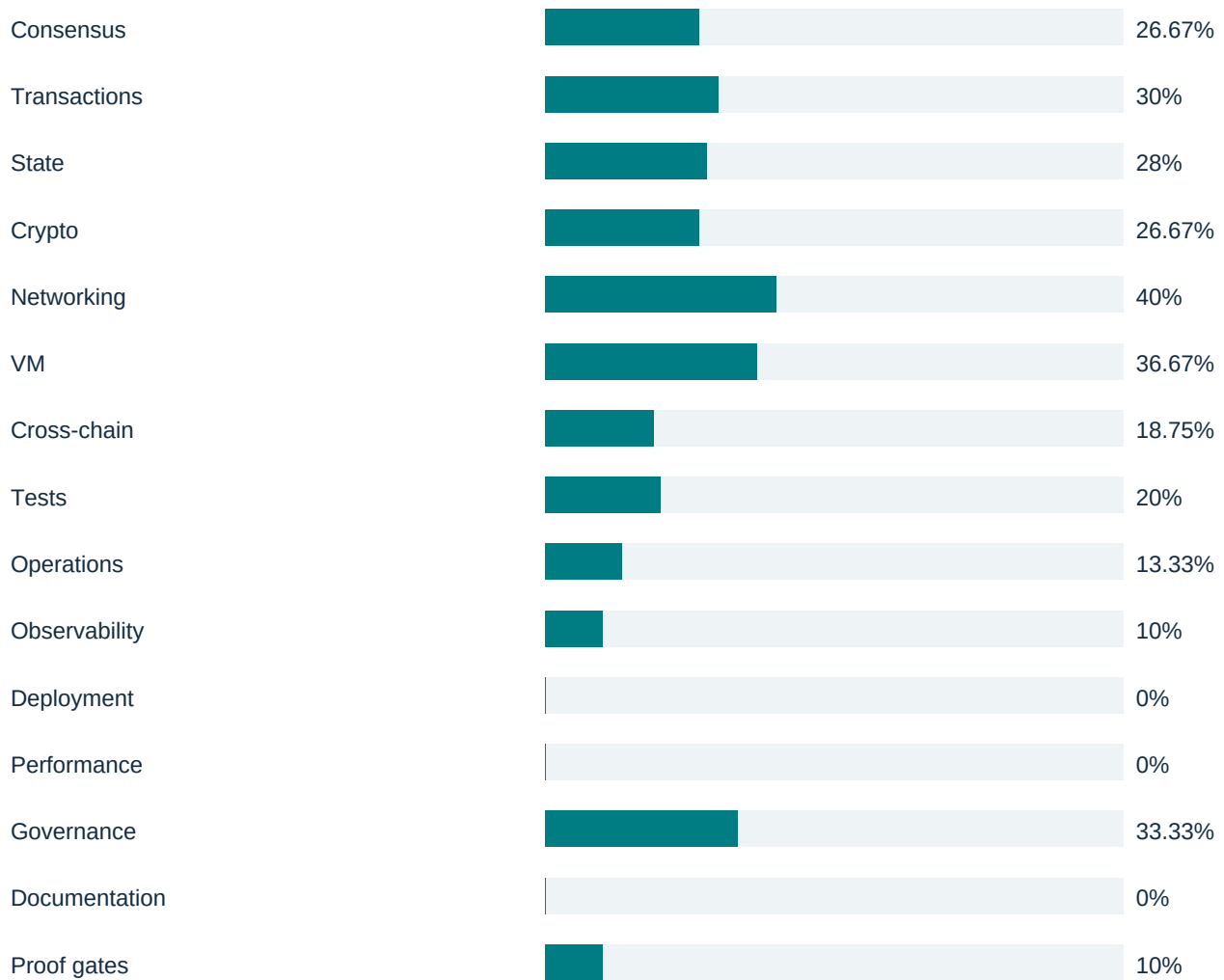


0 at center; rings 20, 40, 60, 80, 100. Overall score is separately safety-capped.

Figure 5 Readiness radar: equal axes, evidence-derived raw scores

Figure 1 shows subsystem evidence score.

Subsystem evidence score

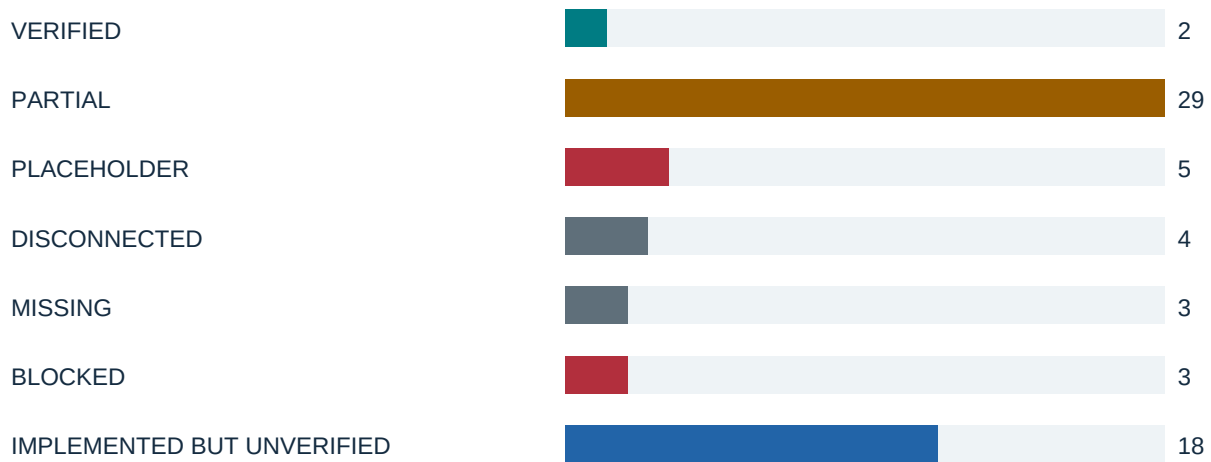


Raw subsystem evidence scores; overall safety cap = 20/100.

Figure 1 Subsystem evidence score

Figure 2 shows feature-status distribution.

Feature-status distribution

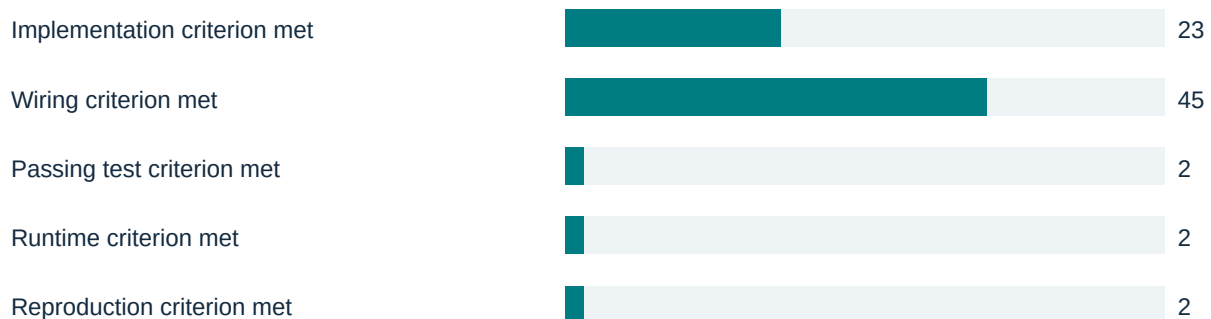


64 scoped capabilities. Status is not code size or deployment coverage.

Figure 2 Feature-status distribution

Figure 4 shows implementation versus verification.

Implementation versus verification

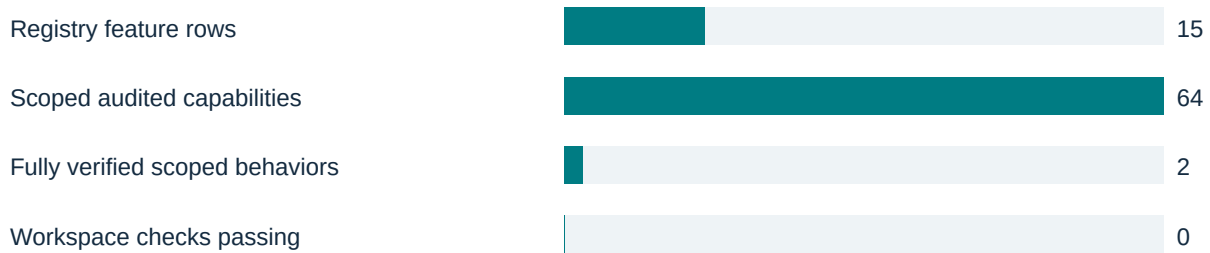


Criteria are deliberately separate; dependency inclusion is not runtime proof.

Figure 4 Implementation versus verification

Figure 25 shows documentation versus executable evidence.

Documentation versus executable evidence

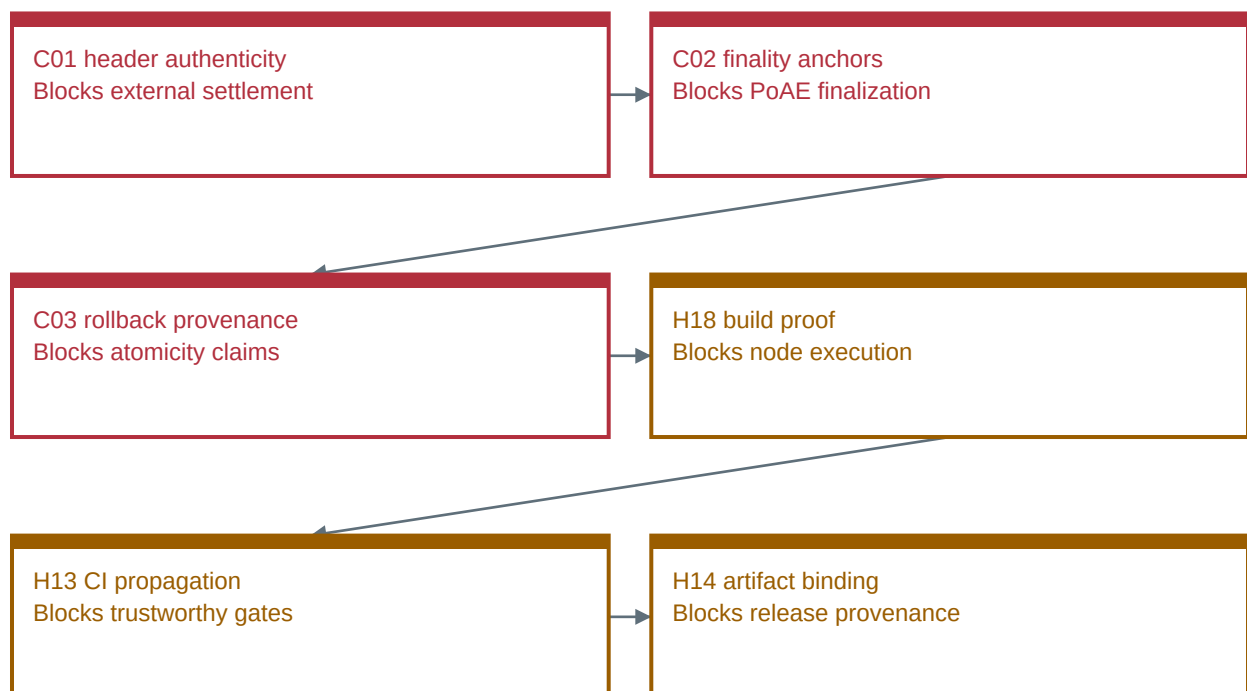


Different scopes shown as counts; registry percentages are not converted into runtime credit.

Figure 25 Documentation versus executable evidence

Figure 26 shows critical dependency coverage.

Critical dependency coverage



I implemented/unverified P partial D disconnected X unsafe V verified N proposed

Qualitative dependency matrix. All six are prerequisites, not measured risk scores.

Figure 26 Critical dependency coverage

Table 7 Feature matrix overview

ID / capability	Status / score	Evidence
FT01 Aura block production	IMPLEMENTED BUT UNVERIFIED / 40%	node/src/service.rs:1059
FT02 GRANDPA finality and fork choice	IMPLEMENTED BUT UNVERIFIED / 40%	node/src/service.rs:601; node/src/service.rs:1136
FT03 Validator rotation	PARTIAL / 20%	pallets/x3-consensus/src/lib.rs:412
FT04 Validator bonded stake and slashing	PARTIAL / 20%	pallets/x3-consensus/src/lib.rs:263
FT05 Flash finality	PARTIAL / 0%	crates/flash-finality/src/lib.rs:389; node/src/service.rs:656
FT06 Genesis construction and live-seed validation	IMPLEMENTED BUT UNVERIFIED / 40%	node/src/chain_spec.rs:760
FT07 Signed native transaction path	IMPLEMENTED BUT UNVERIFIED / 40%	runtime/src/lib.rs:850; node/src/rpc.rs:1050
FT08 Nonce, genesis, era replay checks	IMPLEMENTED BUT UNVERIFIED / 40%	runtime/src/lib.rs:850
FT09 Transaction pool limits and ordering	PARTIAL / 20%	node/src/service.rs:589; node/src/service.rs:371
FT10 Fee charging and refunds	IMPLEMENTED BUT UNVERIFIED / 40%	runtime/src/lib.rs:1091
FT11 Ethereum raw transaction RPC	PARTIAL / 20%	node/src/rpc_frontier.rs:457
FT12 Canonical FRAME storage	IMPLEMENTED BUT UNVERIFIED / 40%	node/src/service.rs:567; runtime/src/lib.rs:834
FT13 Storage upgrade migrations	BLOCKED / 40%	runtime/src/lib.rs:826
FT14 Supply conservation ledger	PARTIAL / 20%	pallets/x3-supply-ledger/src/lib.rs:158
FT15 Snapshot backup and restoration	PARTIAL / 20%	scripts/snapshot-restore.sh:130
FT16 Atomic rollback state provenance	PARTIAL / 20%	pallets/x3-atomic-kernel/src/lib.rs:1030
FT17 Signature primitives and key custody	IMPLEMENTED BUT UNVERIFIED / 40%	node/src/service.rs:323; runtime/src/lib.rs:141
FT18 Dependency advisory handling	PARTIAL / 20%	.cargo/audit.toml:1; evidence/advisories-unfiltered.json
FT19 P2P networking and sync	IMPLEMENTED BUT UNVERIFIED / 40%	node/src/service.rs:881
FT20 Bootstrap helper configuration	DISCONNECTED / 20%	node/src/network.rs:9

ID / capability	Status / score	Evidence
FT21 PoH import verification	PARTIAL / 0%	node/src/service.rs:943
FT22 RPC limiter algorithm (narrow scope)	VERIFIED / 100%	node/src/rpc_middleware.rs:395; evidence/rpc-harness.log
FT23 WASM mini-EVM persistent execution	PARTIAL / 20%	crates/evm-integration/src/mini_evm.rs:102
FT24 SVM execution and account context	PARTIAL / 20%	pallets/x3-kernel/src/wasm_adapters.rs:86
FT25 X3 VM runtime	IMPLEMENTED BUT UNVERIFIED / 40%	pallets/x3-kernel/src/wasm_adapters.rs:182
FT26 X3 Python parser and typechecker	VERIFIED / 100%	x3-lang/cli.py:1; evidence/dsl-tests.log; evidence/dsl-cli.log
FT27 X3 emitter and execution pipeline	PARTIAL / 20%	x3-lang/emitter/x3.py:60
FT28 Rust compiler tracks	IMPLEMENTED BUT UNVERIFIED / 20%	crates/x3-compiler/Cargo.toml:1; x3-lang/compiler/src/lib.rs:1
FT29 Internal cross-VM representation router	IMPLEMENTED BUT UNVERIFIED / 40%	pallets/x3-cross-vm-router/src/lib.rs:1
FT30 Atomic bundle orchestration / Atomic Lock	PARTIAL / 20%	pallets/x3-atomic-kernel/src/lib.rs:839
FT31 Settlement timeout and refund engine	PARTIAL / 20%	pallets/x3-settlement-engine/src/lib.rs:1350
FT32 External header finality oracle	PLACEHOLDER / 20%	pallets/cross-chain-validator/src/lib.rs:332
FT33 Production EVM receipt proof route	PARTIAL / 20%	crates/x3-verification-router/src/evm_receipt.rs:737
FT34 Solana finalized proof route	PLACEHOLDER / 20%	crates/x3-verification-router/src/lib.rs:355
FT35 Validator quorum proof route	PLACEHOLDER / 20%	crates/x3-verification-router/src/lib.rs:323
FT36 Bitcoin SPV route	PARTIAL / 20%	crates/x3-verification-router/src/lib.rs:545
FT37 Bitcoin vault approvals	PLACEHOLDER / 0%	crates/x3-bitcoin-vault/src/lib.rs:324
FT38 Relayer binary delivery	PARTIAL / 20%	crates/x3-relayer/src/main.rs:1442
FT39 Relayer typed submission library	PARTIAL / 20%	crates/x3-relayer/src/submitter.rs:568
FT40 EVM HTLC contract	IMPLEMENTED BUT UNVERIFIED / 20%	X3-contracts/evm/contracts/AtlasHTLC.sol:99

ID / capability	Status / score	Evidence
FT41 SVM HTLC contract	IMPLEMENTED BUT UNVERIFIED / 20%	X3-contracts/svm/programs/x3_htlc/src/lib.rs:1
FT42 External EVM proof verifier contract	PARTIAL / 0%	X3-contracts/evm/contracts/EvmReceiptVerifier.sol:205
FT43 Solver marketplace and intent routing	IMPLEMENTED BUT UNVERIFIED / 20%	crates/x3-intent/Cargo.toml:1; pallets/atomic-trade-engine/src/lib.rs:1
FT44 Validator attestation / proof ledger	PARTIAL / 20%	crates/x3-validator-attestation/src/lib.rs:1; pallets/x3-atomic-kernel/src/lib.rs:1416
FT45 REST/GraphQL gateway	DISCONNECTED / 0%	crates/x3-gateway/src/main.rs:1
FT46 SQL indexer and migrations	DISCONNECTED / 20%	crates/x3-gateway/src/db.rs:1
FT47 Metrics / chain health monitoring	PARTIAL / 20%	node/src/metrics.rs:1; crates/x3-gateway/src/rest.rs:248
FT48 Security and accounting event consumers	DISCONNECTED / 0%	runtime/src/lib.rs:11
FT49 DEX, Forge and LP locker	IMPLEMENTED BUT UNVERIFIED / 40%	pallets/x3-dex/src/lib.rs:1; pallets/x3-token-factory/src/lib.rs:1; pallets/x3-lp-locker/src/lib.rs:1
FT50 Sentinel and economic halt	PARTIAL / 20%	pallets/x3-sentinel/src/lib.rs:1; pallets/x3-invariants/src/lib.rs:618
FT51 Council / treasury / upgrades	IMPLEMENTED BUT UNVERIFIED / 40%	runtime/src/lib.rs:543; runtime/src/lib.rs:826
FT52 Wallet / biometric / recovery pallet	PARTIAL / 20%	pallets/x3-wallet-pallet/src/lib.rs:263
FT53 TypeScript SDK encoding	PLACEHOLDER / 20%	packages/ts-sdk/src/evm.ts:154; packages/ts-sdk/src/svm.ts:230
FT54 Desktop / Tauri OS network integration	PARTIAL / 20%	apps/x3-desktop/src/blockchain/ChainManager.ts:59; apps/tauri-os/package.json:1
FT55 GPU / swarm orchestration	PARTIAL / 0%	node/src/service.rs:1244; crates/x3-swarm-core/Cargo.toml:1
FT56 CI test quality and coverage	PARTIAL / 20%	Makefile:19; node/src/service.rs:2664
FT57 Workspace build / release reproducibility	BLOCKED / 0%	evidence/check-complete.log; rust-toolchain.toml:3
FT58 Fresh-machine bootstrap	BLOCKED / 0%	scripts/fresh_machine_check.sh:5; Makefile:61
FT59 Deployment isolation and validator services	IMPLEMENTED BUT UNVERIFIED / 40%	packaging/systemd/x3-validator.service:1

ID / capability	Status / score	Evidence
FT60 Sustained finalized TPS evidence	MISSING / 0%	crates/x3-bench/src/runner.rs:251
FT61 Mainnet / ProofGate enforcement	PARTIAL / 20%	scripts/mainnet_release_gate.py:84; Makefile:19
FT62 Documentation accuracy / scoreboard	PARTIAL / 0%	CURRENT_MAINNET_STATUS.md:3; FEATURE_REGISTRY.toml:18
FT63 Public testnet recovery drill evidence	MISSING / 0%	node/src/service.rs:2664; scripts/snapshot-restore.sh:130
FT64 Independent launch approval evidence	MISSING / 0%	LAUNCH_SCOPE.md:1

Appendix A expands every scored row with missing work and an acceptance/verification path. The CSV additionally provides all requested business/protocol value, dependencies, security significance, effort and priority fields. Percentages express satisfied evidence criteria, not engineering time remaining. Registry claims are not used as scores.

FINDINGS / FAILURE SCENARIOS / CORRECTIONS

5 Mainnet Blockers

The following mini-briefs are the complete confirmed finding register for this review. Severity ranks the failure consequence under stated preconditions. Static scenarios are not represented as executed attacks. Only H01, H13, H15, H18 and selected Python/gate observations have direct execution evidence. No live funds, signatures or public-network broadcasts were used.

FIELD MANUAL / 05

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

MAINNET KILL LIST: close all Critical and applicable High findings; require a clean full build, real multi-node finalized execution, authenticated external proofs, exact rollback, custody drills and source-bound gate evidence. No waiver is allowed for fund safety, consensus safety, key management or state integrity.

C01 External header validation accepts unproved claims

CRITICAL / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Any signed origin can submit headers. Merkle verification ignores `_expected_root`; quorum is derived from the length of untrusted bytes. Settlement comparison only compares stored fields. `BaseCallFilter` is Everything.

Failure scenario and blast radius: A signed user can poison `LastEvmHeader/LastSvmHeader`, including with a far-future height, defeating the authenticity assumption and blocking later legitimate headers. Fund-release exploitation also depends on settlement-state and secret/leg checks; no live theft was attempted.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Use authenticated chain-specific finalized-header clients with trusted validator/checkpoint transitions, verified signatures, bounded proofs, and authenticated parent links. Reject unsupported chains.

Acceptance criteria / required tests: Arbitrary hashes, duplicate signers, self-selected validator sets, future heights and invalid membership proofs are rejected without writes; canonical external fixtures pass.

Files and lines: `pallets/cross-chain-validator/src/lib.rs:181`; `pallets/cross-chain-validator/src/lib.rs:332`; `pallets/cross-chain-validator/src/lib.rs:279`; `runtime/src/lib.rs:2412`; `runtime/src/lib.rs:980`

Dependencies: None identified; build proof is required before runtime validation. | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p pallet-cross-chain-validator --features std
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

C02 Unsigned finality anchors make certificate checks circular

CRITICAL / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: `record_flash_finality_anchor` accepts any nonzero cert from an unsigned origin; pool checks enforce recency only and propagate it. The first value wins. Finalization later checks equality to that value.

Failure scenario and blast radius: An untrusted transaction can front-run the genuine anchor and permanently poison a block reference. Matching a caller-created anchor cannot prove consensus finality.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Validate an actual GRANDPA justification or authenticated authority certificate bound to block hash, number, set ID and chain ID; remove arbitrary unsigned anchor admission.

Acceptance criteria / required tests: Forged and conflicting anchors cannot enter pool or dispatch; independently verified justification anchors the exact finalized block; late legitimate anchors remain recoverable.

Files and lines: `pallets/x3-atomic-kernel/src/lib.rs:884`; `pallets/x3-atomic-kernel/src/lib.rs:1118`; `pallets/x3-atomic-kernel/src/lib.rs:1416`

Dependencies: None identified; build proof is required before runtime validation. | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p pallet-x3-atomic-kernel --features std
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

C03 Unsigned rollback receipts trust attacker-selected prior state

CRITICAL / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Unsigned `record_leg_execution_receipt` stores `StateDiff` without executor authentication or comparison to actual execution. The configured `CompositeReverter` writes supplied old values into derived VM storage keys.

Failure scenario and blast radius: An attacker can race an executing bundle to install a rollback diff; a later authorized rollback applies attacker-selected VM state. Actual contract impact depends on storage namespace correctness; state provenance is absent even where namespaces are wrong.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Generate diffs inside transactional execution, bind them to `executor/leg/pre-state/access` set, and validate both old and current values. Roll back in reverse dependency order with complete failure propagation.

Acceptance criteria / required tests: Unrelated caller/diff and stale pre-state are rejected; injected rollback failure leaves no partial state; reverse-order replay restores exact pre-root.

Files and lines: `pallets/x3-atomic-kernel/src/lib.rs:1030`; `pallets/x3-atomic-kernel/src/lib.rs:1150`; `pallets/x3-atomic-kernel/src/lib.rs:1289`; `pallets/x3-atomic-kernel/src/vm_revert.rs:342`; `runtime/src/lib.rs:2675`

Dependencies: C02 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p pallet-x3-atomic-kernel --features std
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable `audit-harness/proof` path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H01 Production proof-router alternatives accept arbitrary bytes

HIGH / Confirmed by execution / Public testnet and mainnet

Evidence and root cause: Production-feature adversarial harness observed quorum and Solana verifiers accept a one-byte proof, and legacy EVM verifier accept 64 arbitrary bytes. Quorum and Solana implementations are registered by route type. Gateway proof submission is currently root/half-council gated; legacy EVM is not the selected production EVM route.

Failure scenario and blast radius: Configured routes rely on privileged submitter honesty rather than independently verified deposits. Enabling permissionless relayers before replacement would expose the acceptance flaw directly.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Remove structural verifiers from production and implement signature membership, unique quorum, chain binding and event inclusion; retain governance admission only as defense in depth.

Acceptance criteria / required tests: All three isolated rejection tests pass; actual signed finalized fixtures pass through the runtime gateway, not only the helper.

Files and lines: crates/x3-verification-router/src/lib.rs:323; crates/x3-verification-router/src/lib.rs:355; crates/x3-verification-router/src/lib.rs:276; pallets/x3-crosschain-gateway/src/lib.rs:1162; runtime/src/lib.rs:2479

Dependencies: C01 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test --manifest-path audit-harness/proof/Cargo.toml
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H02 EVM receipt verifier lacks trusted roots and misbinds inclusion

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Verifier reads head height and header from payload, with no authenticated root source. It passes None rather than encoded receipt to the trie verifier and builds a list-wrapped receipt index. The helper uses nonstandard leaf/extension handling.

Failure scenario and blast radius: Real Ethereum proofs can be rejected, and asserted roots/heights cannot establish canonical settlement even if trie parsing succeeds. This is not a working Ethereum light client.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Use canonical typed receipts and RLP(index), verify the exact receipt value under an authenticated receiptsRoot, bind gateway/event/amount/sender/recipient/chain, and verify finality separately.

Acceptance criteria / required tests: Cross-check independent execution-client proof vectors, tampered receipt values and false heads; require both authentic root and exact inclusion.

Files and lines: crates/x3-verification-router/src/evm_receipt.rs:737;
crates/x3-verification-router/src/evm_receipt.rs:593;
crates/x3-verification-router/src/evm_receipt.rs:335

Dependencies: C01 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-verification-router --features production
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H03 Bitcoin verification uses asserted confirmations and incomplete vault approval

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: SPV computes confirmations from caller-supplied chain_tip minus tx_index, rather than authenticated block height. Header PoW exists but no trusted best-work/difficulty chain is established. Vault processing increments confirmations and approvals per call; signer approval stores bytes without signature verification. Router consumes vault constants, not that state machine.

Failure scenario and blast radius: The route lacks authentic Bitcoin chain depth and output/amount/recipient binding. Repeated vault method calls can progress approval without threshold signatures if that library path is used. Direct on-chain reachability of the vault state machine was not established.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Implement persistent best-work header validation with checkpoints/difficulty rules and transaction output proofs; verify distinct authorized signatures over exact deposit/withdrawal payloads.

Acceptance criteria / required tests: Reject invented tips, wrong outputs, wrong recipient/value, repeated approvals and forged signatures; restart and reorg tests preserve accounting.

Files and lines: crates/x3-verification-router/src/lib.rs:557; crates/x3-verification-router/src/lib.rs:617; crates/x3-bitcoin-vault/src/lib.rs:324; crates/x3-bitcoin-vault/src/lib.rs:450

Dependencies: H01 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-bitcoin-vault && cargo test -p x3-verification-router --features production
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H04 Public misbehavior report has no evidence and does not slash currency

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Any signed account may report a registered stake entry with an arbitrary reason. slash_validator updates ValidatorStake metadata, not balances/reserves, and inactive status does not itself remove Aura authority. Existing tests assert this unsupported-report behavior.

Failure scenario and blast radius: A signed caller can repeatedly reduce validator metadata; operators can mistake metadata changes for bonded-asset penalties and removal. This is not proof of theft from Balances.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Route offenses through validated equivocation/unavailability evidence with unique offence IDs; reserve real stake, slash it atomically and coordinate session removal.

Acceptance criteria / required tests: Invalid/replayed evidence cannot slash; total reserved balances and treasury/burn conservation reconcile after each valid offence; removed keys cannot author/vote.

Files and lines: pallets/x3-consensus/src/lib.rs:263; pallets/x3-consensus/src/lib.rs:381; pallets/x3-consensus/src/tests/slashing.rs:31; runtime/src/lib.rs:1063

Dependencies: None identified; build proof is required before runtime validation. | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p pallet-x3-consensus --features std
```

Verification note: this command is a remediation validation entryptpoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entryptpoints disabled if rollback cannot preserve state.

H05 Validator rotation ignores requested activation delay at session boundary

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: `set_validators` stores a future activation block, but `SessionManager::new_session` returns `NextValidators` without checking it; `start_session` clears the pending set.

Failure scenario and blast radius: A governance change scheduled beyond the next session can enter the session pipeline prematurely, and local validator metadata can diverge from session authorities.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Use one activation state machine for queued keys, session selection and metadata. Reject empty/duplicate/unkeyed sets and enforce the same scheduled boundary.

Acceptance criteria / required tests: A change delayed over several sessions is not activated early; restart and session transition preserve identical authority sets on all nodes.

Files and lines: `pallets/x3-consensus/src/lib.rs:229`; `pallets/x3-consensus/src/lib.rs:412`; `runtime/src/lib.rs:1042`

Dependencies: H04 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p pallet-x3-consensus --features std
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable `audit-harness/proof` path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H06 Flash-finality opt-in disables GRANDPA without a proven replacement

HIGH / Confirmed by static inspection / Experimental finality; public testnet and mainnet if enabled

Evidence and root cause: Flash flag disables GRANDPA even when no flsh key is available. Flash vote verification checks signer self-signature but no configured authority membership. The named voter consumes finality notifications rather than independently finalizing blocks.

Failure scenario and blast radius: Opt-in can stall finality; self-created keys can contribute votes in the gadget. Default GRANDPA path is separate and is not alleged broken by this finding.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Keep GRANDPA mandatory until a separately audited finality engine proves authority membership, safety and liveness; reject incompatible flags or missing keys at startup.

Acceptance criteria / required tests: Wrong keys/sets cannot form certificates; missing keys fail startup; partition, equivocation and recovery tests establish one finalized history.

Files and lines: node/src/service.rs:656; node/src/service.rs:962; node/src/service.rs:1937; crates/flash-finality/src/lib.rs:389

Dependencies: None identified; build proof is required before runtime validation. | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-chain-node --lib
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H07 Default mini-EVM executes against disposable synthetic state

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Each call creates MemoryBackend accounts, seeds caller with u128::MAX, installs payload as code, and returns empty state_changes. The adapter passes zero caller/value. Its state_root hashes selected inputs/output rather than persisted account state.

Failure scenario and blast radius: Successful opcode execution does not demonstrate persistent EVM contracts, authenticated caller semantics or real gas payment. Internal representation transfers must not be marketed as full EVM interoperability.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Connect actual runtime state, caller/value, logs, state commitment and gas charging through a transactional interpreter; run canonical EVM conformance tests.

Acceptance criteria / required tests: A contract SSTORE persists across blocks/restart, a second user cannot impersonate the first, out-of-gas fully reverts and fees reconcile.

Files and lines: crates/evm-integration/src/mini_evm.rs:102;
crates/evm-integration/src/mini_evm.rs:148; pallets/x3-kernel/src/wasm_adapters.rs:34;
runtime/src/lib.rs:1564

Dependencies: C03 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-chain-runtime --features frontier
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H08 Ethereum submission RPC bypasses the transaction-pool lifecycle

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: `eth_sendRawTransaction` calls a runtime API directly. The runtime function expects a custom caller/to/value/data frame, not a signed Ethereum envelope, and no pool submission appears in this handler. Default feature path reports EVM disabled.

Failure scenario and blast radius: When frontier is enabled, the method cannot establish signed admission, durable inclusion or finality by returning a hash from a runtime query. The API cannot be treated as Ethereum-compatible submission.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Decode and verify canonical signed transactions and submit an appropriate runtime extrinsic through the pool; report receipts only after inclusion and finality.

Acceptance criteria / required tests: Submit a signed transfer over RPC, observe pool admission, block inclusion, finality and persisted balance change on another node; reject replay/wrong chain ID.

Files and lines: `node/src/rpc_frontier.rs:457`; `runtime/src/lib.rs:3425`; `node/src/rpc.rs:341`

Dependencies: H07 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-chain-node --features frontier --lib
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H09 Relay manual extrinsic encoding and signing do not match runtime

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Builder uses 0x81, missing address/signature enum tags and full pallet index, two-byte immortal era, zero genesis/block hashes and an Alice fallback. It omits signed-extension context. Gateway submit origin is governance, not an arbitrary relayer.

Failure scenario and blast radius: Proof helpers can produce bytes that cannot be admitted as valid runtime transactions; dependency wiring alone does not make the relayer operational.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Use metadata-derived typed calls and the exact signed extensions, chain/genesis versions and custody signer; introduce an explicit governed relayer authorization model.

Acceptance criteria / required tests: Exact production runtime decodes, authenticates, includes and finalizes generated deposit/release transactions; no fallback signer or zero-chain context.

Files and lines: crates/x3-relayer/src/submitter.rs:437; crates/x3-relayer/src/submitter.rs:568; runtime/src/lib.rs:850

Dependencies: H01, H08 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-relayer
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H10 Relay checkpoints can skip failed events and are not durable

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Binary stores processed events/cursors in HashMaps. `db_path` is declared but no persistence path was found in this binary. `max_block_seen` advances before submission succeeds and is saved after failures.

Failure scenario and blast radius: A failed earlier event can fall behind a later cursor; restart loses deduplication and causes unbounded rescans. Event observation is not finalized delivery.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Persist transactional inbox/outbox and per-event delivery state; checkpoint only a contiguous completed finalized range with reorg rollback and idempotent retries.

Acceptance criteria / required tests: Fail submission for an early event, process later events, restart and recover the failed event exactly once; bound RPC scans and memory.

Files and lines: `crates/x3-relayer/src/main.rs:535`; `crates/x3-relayer/src/main.rs:795`;
`crates/x3-relayer/src/main.rs:827`

Dependencies: H09 | Priority: P0 | Owner: Protocol security | Complexity: L (ordinal estimate).

```
cargo test -p x3-relayer
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H11 Gateway binary never starts its REST/GraphQL implementation

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: The only binary main initializes tracing, logs and exits. There is no lib.rs; main declares none of the database/router modules. API source and dependency comments do not place handlers in the executable.

Failure scenario and blast radius: A successfully built gateway can exit without serving any route. Claimed bridge and database integration through gateway dependencies is disconnected.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Implement validated configuration loading, database migrations, cache setup, router binding and graceful shutdown; expose dependency readiness separately from liveness.

Acceptance criteria / required tests: Start the built binary against disposable services, serve actual API/GraphQL data, fail readiness when DB is down, and recover after restart.

Files and lines: crates/x3-gateway/src/main.rs:1; crates/x3-gateway/Cargo.toml:12;
crates/x3-gateway/src/rest.rs:85

Dependencies: None identified; build proof is required before runtime validation. | Priority: P1 | Owner: Backend / SRE | Complexity: L (ordinal estimate).

```
cargo test -p x3-gateway && cargo run -p x3-gateway
```

Verification note: this command is a remediation validation endpoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected endpoints disabled if rollback cannot preserve state.

H12 SDK EVM and SVM encoding contains production placeholders

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Address derivation slices public-key bytes, functionSelector uses a 32-bit rolling hash, and SVM instruction encoding writes zero for all account/program indices.

Failure scenario and blast radius: Users can derive the wrong address or send wrong function/account encodings. A working transport does not make this SDK protocol compatible.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Use canonical Keccak address/selector derivation and canonical Solana message compilation with real account metas and program indices.

Acceptance criteria / required tests: Match independent reference vectors for EVM selectors/addresses and multi-account SVM messages, then finalize one transaction per advertised protocol.

Files and lines: packages/ts-sdk/src/evm.ts:140; packages/ts-sdk/src/evm.ts:154;
packages/ts-sdk/src/evm.ts:440; packages/ts-sdk/src/svm.ts:230

Dependencies: None identified; build proof is required before runtime validation. | Priority: P1 | Owner: SDK engineer | Complexity: L (ordinal estimate).

```
npm --prefix packages/ts-sdk test -- --runInBand
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H13 CI make recipes swallow failures and reference missing targets

HIGH / Confirmed by execution / Public testnet and mainnet

Evidence and root cause: Observed make test-x3-wallet exit 0 while Cargo errored. Recipes pipe into tail under the default shell without pipefail. test-node-build has no rule; -p node is not the x3-chain-node package.

Failure scenario and blast radius: Required checks can claim success without testing the intended pallet; another gate fails before assessing the node.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Preserve exit codes, use exact package names and define/reconcile workflow targets; test gates against a deliberately failing test in disposable fixtures.

Acceptance criteria / required tests: Cargo failure produces nonzero make/CI exit; every workflow command resolves; mainnet-required aggregate cannot pass with skipped/missing jobs.

Files and lines: Makefile:19; Makefile:29; Makefile:38; .github/workflows/production-gate.yml:25

Dependencies: None identified; build proof is required before runtime validation. | Priority: P0 | Owner: Release engineering | Complexity: S (ordinal estimate).

```
make test-x3-wallet; make -n test-node-build
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H14 Release gate accepts stale build artifacts without source binding

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: `check_build` skips compilation if a path exists; `genesis` checks `parse JSON` and search `production_config` text. No binary/source hash or generated-spec equality is required here. Other provenance workflows do not repair this local decision.

Failure scenario and blast radius: An old binary or a parseable but wrong `genesis` can satisfy these subchecks for new source.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Build from a clean locked commit, verify compiler/runtime/features, regenerate `genesis`, sign a source-bound manifest and reject dirty or stale evidence.

Acceptance criteria / required tests: Mutating source or substituting a binary/spec invalidates evidence and blocks release; two builders reproduce the same WASM.

Files and lines: `scripts/mainnet_release_gate.py:84`; `scripts/mainnet_release_gate.py:118`

Dependencies: H13 | Priority: P0 | Owner: Release engineering | Complexity: M (ordinal estimate).

```
python3 scripts/mainnet_release_gate.py
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable `audit-harness/proof` path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H15 Dependency audit pass masks unresolved advisory matches

HIGH / Confirmed by execution / Public testnet and mainnet

Evidence and root cause: Configured offline scan reports zero unsuppressed vulnerabilities with 35 ignored advisory IDs. Same lockfile outside repository config reports 53 advisory/package-version matches, 18 unmaintained and 13 unsound warnings. Cached DB freshness is unknown.

Failure scenario and blast radius: A green configured audit is not proof that dependencies are unaffected. Reachability and platform-specific exploitability require per-binary analysis; 53 matches are not 53 confirmed exploitable node vulnerabilities.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Triage each advisory against shipped binaries/features; upgrade affected paths and narrowly justify time-limited exceptions with technical evidence. Verify fresh official advisory data before release.

Acceptance criteria / required tests: No reachable unmitigated critical/high advisories; exceptions have owners, expiry, exact dependency paths and regression evidence.

Files and lines: .cargo/audit.toml:1; Cargo.lock:1; evidence/advisories-unfiltered.json

Dependencies: H13 | Priority: P0 | Owner: Security / dependency engineering | Complexity: L (ordinal estimate).

```
cargo audit --no-fetch --stale --no-yanked --file Cargo.lock --json
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H16 Supply invariant finalization work is unbounded by asset count

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: `on_finalize` iterates all asset ledgers and builds proof vectors; the hook has no measured per-asset `on_initialize` reservation in this implementation. Historical block retention does not bound per-block asset work.

Failure scenario and blast radius: As asset count grows, mandatory work can exceed block resources and undermine liveness independently of transaction admission limits.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Bound asset cardinality/work or use an incremental authenticated aggregate; benchmark and reserve hook weight with proof size accounting.

Acceptance criteria / required tests: Worst-case asset count fits declared block budget; adversarial asset creation cannot make next-block mandatory work unbounded.

Files and lines: `pallets/x3-supply-ledger/src/lib.rs:158`; `pallets/x3-supply-ledger/src/lib.rs:166`

Dependencies: C01 | Priority: P0 | Owner: Runtime performance | Complexity: L (ordinal estimate).

```
cargo test -p pallet-x3-supply-ledger --features std
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable `audit-harness/proof` path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H17 EVM external verifier is blocked in production and bypassable by mode

HIGH / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Production mode unconditionally reverts signature verification. Owner can enable structural-only testnet mode. Quorum is only constrained to 1..N, not the documented supermajority.

Failure scenario and blast radius: Bridge either cannot verify withdrawals or relies on unverified signature slots if the owner enables test mode. Deployment and owner state were not queried.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Deploy a real supported verifier with domain/set binding and unique quorum; remove bypass from funded deployments and govern rotations with delays.

Acceptance criteria / required tests: Real proof accepted, zero/duplicate/forged signatures rejected; funded deployment cannot enable structural bypass; owner rotation preserves trust policy.

Files and lines: X3-contracts/evm/contracts/EvmReceiptVerifier.sol:205;
X3-contracts/evm/contracts/EvmReceiptVerifier.sol:183;
X3-contracts/evm/contracts/EvmReceiptVerifier.sol:73

Dependencies: H01 | Priority: P0 | Owner: Smart-contract security | Complexity: L (ordinal estimate).

```
forge test --root X3-contracts/evm
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M01 Readiness documentation contradicts executable evidence

MEDIUM / Confirmed by execution / Public testnet and mainnet

Evidence and root cause: Status document simultaneously claims clean compilation and a pre-existing compile error. Readiness checker passes path/mode consistency; this is not runtime validation or a recalculation from execution proofs.

Failure scenario and blast radius: Sponsors/operators can infer maturity from scores that do not verify the promised behavior.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Generate status from commit-bound feature proof criteria and preserve blocked/failed results. Remove operational claims that only have path-existence evidence.

Acceptance criteria / required tests: Any failed core proof lowers its status; stale commit evidence is rejected; readiness prose and matrix derive from the same data.

Files and lines: CURRENT_MAINNET_STATUS.md:3; CURRENT_MAINNET_STATUS.md:63; FEATURE_REGISTRY.toml:18; evidence/consistency.log

Dependencies: None identified; build proof is required before runtime validation. | Priority: P2 | Owner: Release / technical writing | Complexity: S (ordinal estimate).

```
bash scripts/check-readiness-consistency.sh
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M02 Finality and multi-node proof tests are ignored

MEDIUM / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Service-level import, finalized-head and two-validator tests carry ignore attributes; two-node reason explicitly cites stalled networking/finality shutdown. An active HTTP new-head test is not finality proof.

Failure scenario and blast radius: Ordinary workspace green results can exclude the tests that establish cross-node finalized state.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Run each node in a separate process with isolated logger/ports/data; hard-gate finalized transaction and reconnect recovery tests.

Acceptance criteria / required tests: At least four validators agree on finalized hash/root through restart and one-node failure; ignored critical tests count as release failure.

Files and lines: node/src/service.rs:2429; node/src/service.rs:2580; node/src/service.rs:2664

Dependencies: H13 | Priority: P1 | Owner: Distributed-systems QA | Complexity: L (ordinal estimate).

```
cargo test -p x3-chain-node two_validator_nodes_submit_on_first_observe_finalized_bridge_state_on_second
-- --ignored --nocapture
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M03 Gateway health and event spines do not prove dependencies

MEDIUM / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Health returns HTTP 200 unconditionally. Security and accounting hooks log dropped events; names containing FailClosed do not enforce a state rollback or durable delivery. Gateway health is currently disconnected with the binary.

Failure scenario and blast radius: Operators can observe liveness while indexing/accounting subscribers are absent; incident evidence may be lost.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Separate liveness/readiness, probe DB/indexer lag and durable event delivery, and make missing required consumers explicit startup/readiness failures.

Acceptance criteria / required tests: Stop database/event sink and see readiness fail; replay missed events exactly once after recovery.

Files and lines: crates/x3-gateway/src/rest.rs:248; runtime/src/lib.rs:11; runtime/src/lib.rs:24

Dependencies: H11 | Priority: P2 | Owner: Observability / SRE | Complexity: L (ordinal estimate).

```
cargo test -p x3-gateway
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M04 Restore validates archive shape, not trusted checksum or chain state

MEDIUM / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Restore checks tar readability then extracts; it does not verify the separately written SHA-256 manifest, chain identity, state root or binary/schema compatibility.

Failure scenario and blast radius: A corrupted or substituted readable archive can be accepted; successful extraction is not successful chain recovery.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Verify authenticated manifest, confined archive paths, chain/genesis/schema identity and finalized state before activation.

Acceptance criteria / required tests: Tampered archive fails before extraction; restored node catches up and matches independently recorded finalized state.

Files and lines: scripts/snapshot-restore.sh:130; scripts/snapshot-restore.sh:141

Dependencies: H14 | Priority: P2 | Owner: SRE / storage | Complexity: L (ordinal estimate).

```
bash -n scripts/snapshot-restore.sh
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M05 Python DSL emitter tests fail under the current proof contract

MEDIUM / Confirmed by execution / Public testnet and mainnet

Evidence and root cause: Four selected emitter tests fail with ProofRequiredError; six parser/typechecker/simulator checks pass. The failing implementation rejects missing proof material rather than silently manufacturing it.

Failure scenario and blast radius: The published happy path and fixtures are not integrated with the enforced proof requirement; simulated planning is not execution.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Supply real verification fixtures and reconcile emitters, runner and CLI contracts without weakening fail-closed proof checks.

Acceptance criteria / required tests: All emitter tests pass with independently validated proof fixtures and reject empty/tampered bundles; one local end-to-end emission is consumed by the real runtime.

Files and lines: x3-lang/emitter/x3.py:60; x3-lang/tests/test_emitter.py:1; evidence/dsl-tests.log

Dependencies: H01 | Priority: P1 | Owner: Language / SDK | Complexity: L (ordinal estimate).

```
/usr/bin/python3 -m pytest -q x3-lang/tests/test_emitter.py
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M06 Desktop defaults select an unregistered chain

MEDIUM / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: initialize registers x3-local and ethereum, while getActiveAdapter defaults activeChainId to x3-testnet. Config also labels a remote HTTP endpoint X3 Local Dev.

Failure scenario and blast radius: Default adapter lookup can fail after initialization; network identity presented to users is ambiguous.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Register configured enabled adapters and select an initialized network by verified chain/genesis identity; use secure transport for public endpoints.

Acceptance criteria / required tests: Default startup obtains the intended adapter, missing network fails clearly, and wrong genesis cannot be signed against.

Files and lines: apps/x3-desktop/src/blockchain/ChainManager.ts:59;
apps/x3-desktop/src/blockchain/ChainManager.ts:117

Dependencies: None identified; build proof is required before runtime validation. | Priority: P2 | Owner: Desktop / wallet | Complexity: S (ordinal estimate).

```
npm --prefix apps/x3-desktop test
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

M07 Optimizer throughput is not network TPS evidence

MEDIUM / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Runner computes tx/sec from number of compiler samples divided by compile time. It measures compiler/optimizer throughput, not mempool-to-finality transaction rate.

Failure scenario and blast radius: Reusing these numbers as blockchain TPS would mislead capacity planning and sponsor claims.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Keep compiler metrics labeled; add finalized unique transaction load tests with latency histograms and resource/recovery evidence.

Acceptance criteria / required tests: Publish reproducible sustained finalized TPS, rejection rate, p50/p95/p99 finality latency and hardware/profile/commit details; no synthetic metric substitution.

Files and lines: crates/x3-bench/src/main.rs:3; crates/x3-bench/src/runner.rs:251

Dependencies: None identified; build proof is required before runtime validation. | Priority: P3 | Owner: Performance engineering | Complexity: M (ordinal estimate).

```
cargo run -p x3-bench -- --baseline
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

L01 Fresh-machine helper does not verify the required environment

LOW / Confirmed by static inspection / Public testnet and mainnet

Evidence and root cause: Helper hard-requires python alias and permits missing cargo/node via `|| true`. The Makefile fresh-machine path uses package node. Audit environment lacks python/pnpm/forge/anchor on PATH; isolated JS copy has no installed dependencies.

Failure scenario and blast radius: Fresh-machine success/failure is not a reliable installation assessment; commands need one pinned, reproducible setup contract.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Validate exact tools/versions, native libraries and lockfile installs; distinguish unavailable tool from failed project test.

Acceptance criteria / required tests: Clean image builds all intended targets and runs real smoke tests with explicit offline/cache prerequisites.

Files and lines: `scripts/fresh_machine_check.sh`:5; `Makefile`:61; `rust-toolchain.toml`:3

Dependencies: None identified; build proof is required before runtime validation. | Priority: P2 | Owner: Developer experience | Complexity: S (ordinal estimate).

```
bash scripts/fresh_machine_check.sh
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

H18 Workspace verification fails in WASM runtime build

HIGH / Confirmed by execution / Public testnet and mainnet

Evidence and root cause: Continued cargo check, cargo test and cargo clippy each exit 101. crypto-common compilation reports E0152 duplicate sized lang item because build-std and installed WASM core artifacts are both loaded. Initial short runs timed out during compilation; final runs establish failure.

Failure scenario and blast radius: No complete node/runtime build or workspace test pass was demonstrated for this source state in the audit environment. Existing binaries and historic reports cannot supply that proof.

Immediate containment: Keep the affected path disabled and do not authorize real-value use until its acceptance criteria pass.

Production correction: Reconcile pinned rust-src/target/build-std configuration and dependency build graph in a fresh build directory; preserve both native and WASM validation without skip-WASM workarounds.

Acceptance criteria / required tests: From the frozen source state, all required workspace checks and release/testnet feature builds exit zero and produce source-bound artifacts.

Files and lines: runtime/build.rs:1; rust-toolchain.toml:3; evidence/check-complete.log; evidence/test-complete.log; evidence/clippy-complete.log

Dependencies: None identified; build proof is required before runtime validation. | Priority: P1 | Owner: Rust build / release engineering | Complexity: M (ordinal estimate).

```
cargo check --workspace && cargo test --workspace && cargo clippy --workspace --all-targets -- -D warnings
```

Verification note: this command is a remediation validation entrypoint, not a claim that the missing adversarial tests already exist. Add the named acceptance cases and retain logs, roots, receipts and source hash. For H01 use the durable audit-harness/proof path in this artifact directory.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

ASSETS / ATTACKERS / CONTROLS

6 Security Threat Model

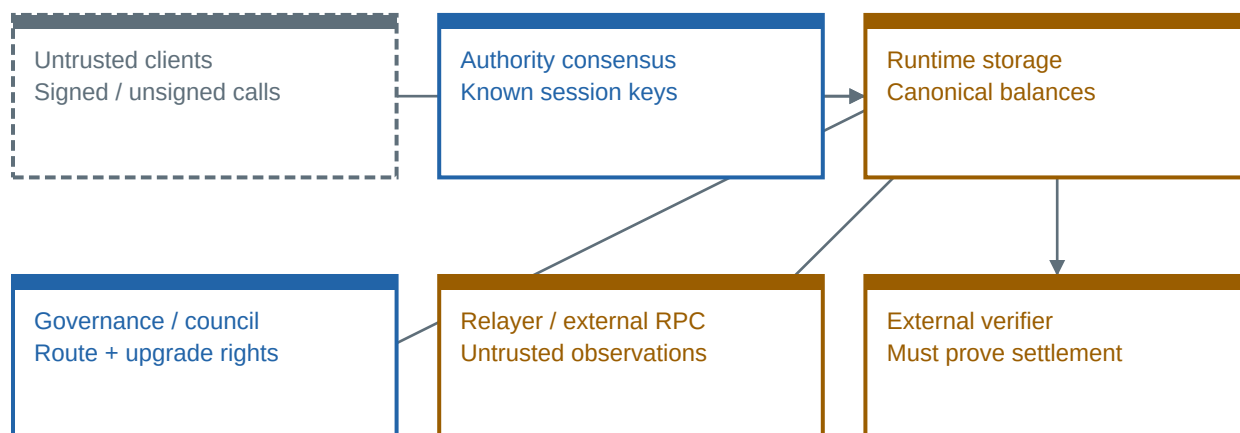
This is a repository-specific engineering threat model, not an independent formal security audit. Protected assets include canonical supply, user escrow, validator authority, proof authenticity, finalized history, wallet/custody keys, release artifacts and operator evidence. Likely adversaries include arbitrary RPC clients, malicious peers, compromised relayers, dishonest validators, compromised governance/contract owners and dependency or deployment attackers.

FIELD MANUAL / 06

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Figure 10 shows trust boundaries.

Trust boundaries

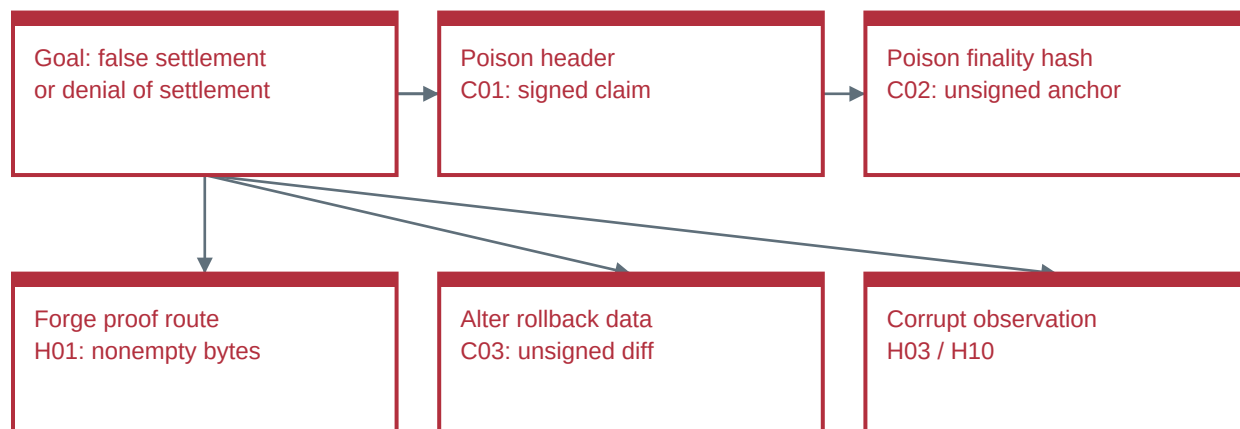


I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Static wiring; no live deployment is asserted.

Figure 10 Trust boundaries

Figure 18 shows attack tree: false settlement.

Attack tree: false settlement



I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Attack hypotheses tied to findings; fund-theft end-to-end exploitation was not performed.

Figure 18 Attack tree: false settlement

Figure 6 shows risk heatmap: finding counts by inferred likelihood and impact, not probabilities.

Risk heatmap — ordinal review estimates

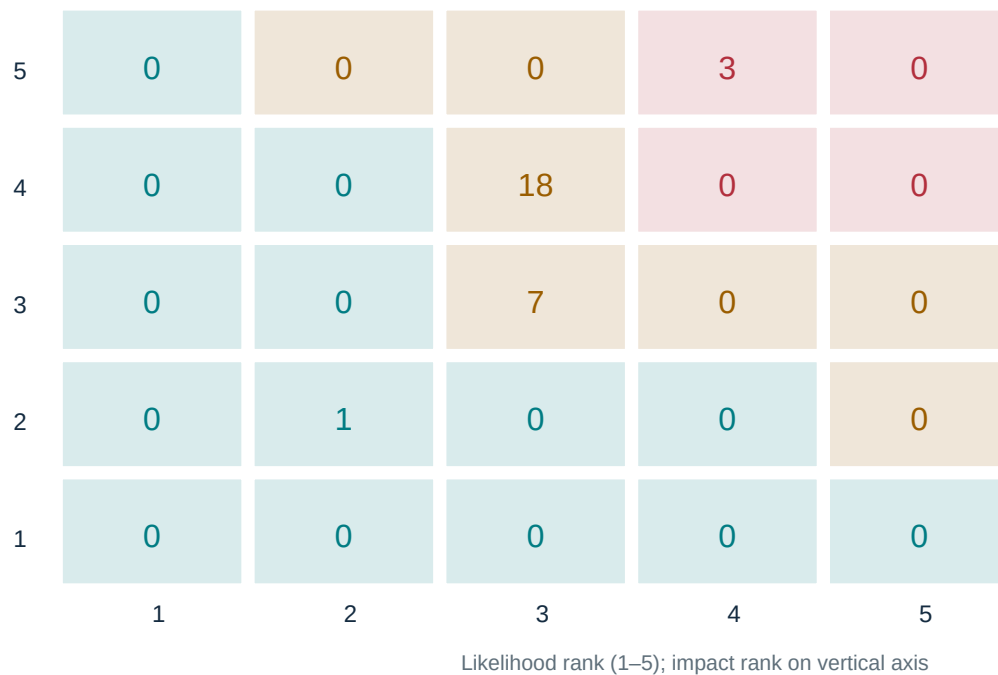
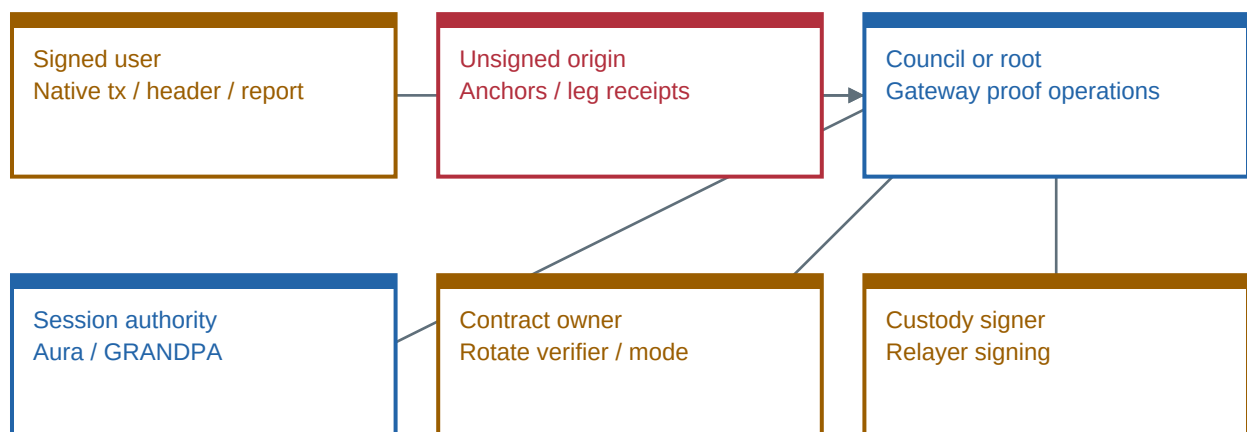


Figure 6 Risk heatmap: finding counts by inferred likelihood and impact, not probabilities

Figure 19 shows privilege map.

Privilege map



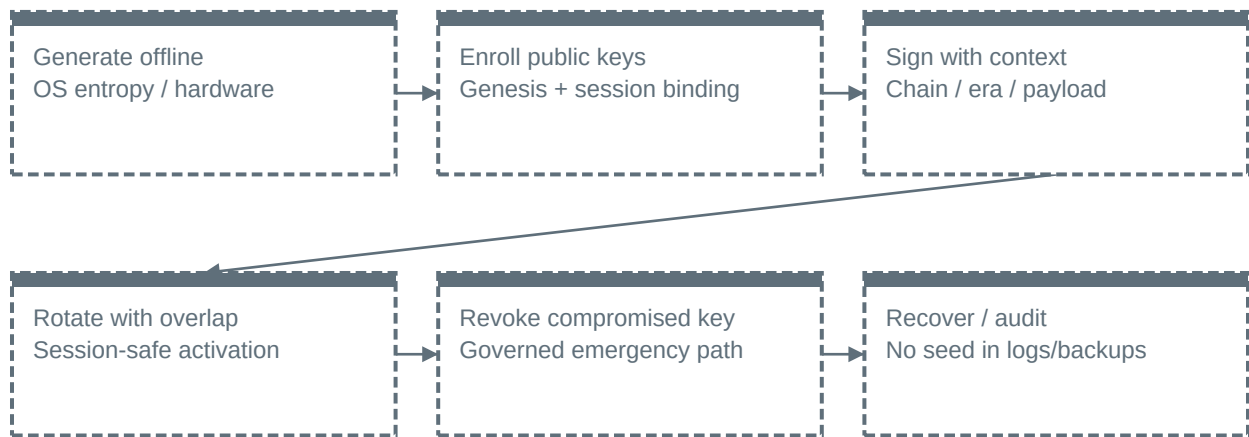
I implemented/unverified P partial D disconnected X unsafe V verified N proposed

Static wiring; no live deployment is asserted.

Figure 19 Privilege map

Figure 20 shows key lifecycle acceptance plan.

Key lifecycle acceptance plan



I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Required lifecycle. Secure custody and recovery were not demonstrated in this audit.

Figure 20 Key lifecycle acceptance plan

Table 8 Security controls and remaining exposure

Boundary	Existing control	Unmitigated risk / required test
Client → runtime	SCALE signatures + nonce/genesis/era/weight	Custom Ethereum RPC/SDK encoders diverge (H08/H12); test canonical signed wire formats.
Unsigned → atomic kernel	Nonzero values, bundle state and recency	No authenticated finality/diff provenance (C02/C03); unauthorized pool and dispatch tests.
External chain → header store	Structural checks and byte counts	No authenticated roots or validators (C01); real negative proof vectors.
Relayer → gateway	Root/half-council origin	Privileged trust hides broken verifier behavior (H01-H03); proof verification must remain independent.
Validator → consensus	Standard Aura/GRANDPA wiring	Custom offence and rotation defects (H04/H05); bond/session tests.
Owner → EVM verifier	Ownable administration	Structural mode bypass and incomplete verification (H17); cannot permit on funded deployment.
Code → release	CI workflows and gates	Suppressed failures/stale artifacts/advisories (H13-H15); mutation tests for gates.
Process → recovery	systemd hardening, archive helper	No observed restore drill; authenticated manifest and state-root verification required.

Key lifecycle evidence is limited to source wiring: local keystore insertion, configured authority keys, signing libraries and custody-related code. No production key material was read or exported. Development seeds must never silently substitute for custody identities. The relayer builder fallback is a concrete violation (H09). Known test identities in source are development fixtures, not evidence of protected production custody.

INVARIANTS / CONSERVATION / RECOVERY

7 Consensus, Transactions and State Integrity

The standard consensus substrate is real code, but project-specific integrations determine whether the system is safe. Block inclusion and finality are separate observations. A local event or a returned transaction hash is not a finalized receipt.

FIELD MANUAL / 07

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Table 9 Protocol invariant register

Invariant	Enforcement evidence	Failure and missing proof
One finalized history	Aura import + GRANDPA voter, service.rs:601/:1136	Run partition, equivocation, restart and catch-up; ignored finality tests do not satisfy it.
Only authorized validator changes	Root set_validators; runtime SessionManager=X3Consensus	Activation delay ignored by new_session (H05); prove keys and session transition together.
Penalty requires an offence	OnOffenceHandler exists	Public report lacks evidence; metadata slash is not currency slash (H04).
Replay cannot debit twice	SignedExtra includes genesis/era/nonce	Run duplicates/concurrent nonces through real RPC/pool/finality.
Fees conserve balances	ChargeTransactionPayment and CurrencyAdapter	Prove treasury/burn/refund deltas on success and failure with exact runtime.
Represented supply $\leq$ canonical supply	Supply ledger checked arithmetic and on_finalize verification	Bound work (H16); authenticate every credited external leg (C01/H01).
Atomic failure restores pre-state	Storage-layer transactions and CompositeReverter	Untrusted prior-state diffs break provenance (C03); restore exact root under induced failure.
No fabricated execution/finality proof	Atomic finalization compares anchor and commitment	Anchor is attacker-submittable (C02); nonzero values/public metadata are not proofs.
Timed-out escrow is recoverable	Settlement timeout/claim/refund branches	Verify timeout unit conversions, exact boundaries, one-time refund and failure after every leg.
Upgrades preserve state meaning	Executive migration tuple	No successful old-schema migration or restoration execution in this audit.

Genesis constructors distinguish development and live configurations and reject several missing live inputs. Public setup still needs a ceremony with real operator public keys, verified authority uniqueness, council/treasury separation, bootnode identities, chain ID and exact WASM hash. Generated or historical JSON alone is insufficient. Raw and plain chain specs must encode the same intended state.

State is committed through the Substrate backend and runtime execution. Gateway PostgreSQL data is derived/indexed data, not the consensus authority. Reconstructing an index is different from restoring validator state; both need drills. Reorganization handling must distinguish best-head observations from finalized events, including relayer cursor rollback.

ATOMICITY / FINALITY / AUTHENTICATION

8 Smart Contracts, VM and Cross-Chain Systems

Internal representation movement, VM opcode execution and asynchronous cross-chain settlement are different capabilities. An atomic FRAME storage transaction can restore local state; it cannot synchronously undo an already-finalized transfer on an external chain. Cross-chain HTLC safety also depends on authenticated observation, timeout ordering, replay protection and eventual refund availability.

FIELD MANUAL / 08

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Table 10 Trustless, trusted or unsupported claim?

Capability	Actual trust / status	Evidence
Internal representation router	Implemented, runtime proof outstanding	pallets/x3-cross-vm-router; local ledger conservation is not external asset proof.
Mini-EVM adapter	Disposable interpreter state; incomplete production execution	H07: synthetic account funding/zero caller and no persisted state changes.
SVM / X3 adapters	Partial contexts / unverified runtime semantics	wasm_adapters.rs:86/:182; prove account state and native/WASM parity.
Quorum / Solana proof routes	Structural acceptance, currently privileged gateway admission	H01 confirmed arbitrary-byte acceptance.
EVM receipt route	Partial inclusion implementation, unauthenticated head/root	H02; canonical independent fixtures required.
Bitcoin route	Header PoW helper present, complete chain/output proof absent	H03; asserted depth and missing asset binding.
EVM external verifier	Production mode fails closed; owner can enable unverified structural mode	H17; not an operating trustless bridge.
EVM/SVM HTLC contracts	Real source, deployment/runtime proof unavailable	Foundry/Anchor tools absent; no on-chain state checked.
Atomic proof ledger	PoAE records are not independently authenticated finality/execution	C02/C03 plus public-field receipt commitment.

Contract review priorities include reentrancy, access control, pause/rotation rights, fee-on-transfer token accounting, refunds at the exact expiry boundary, duplicate claims, integer limits and mismatched chain/domain encodings. AtlasHTLC uses SafeERC20 and ReentrancyGuard in source, but receives a nominal token amount without an observed balance-delta proof. That is a review/test requirement here, not a demonstrated exploit finding. SVM programs need account ownership, signer/PDA constraints, CPI restrictions and proof-instruction binding checks under a local validator.

The Ethereum settlement helper separately derives a block number from tx_hash bytes and equates a transaction hash to a receipt hash (pallets/x3-settlement-engine/src/lib.rs:2124). Those are not canonical Ethereum identifiers. This corroborates C01/H02 interoperability failure, rather than adding duplicate severity counts. Ethereum transaction/receipt conventions are referenced in Appendix E.

MEASURED / UNMEASURED / TEST DESIGN

9 Performance and Scaling

No sustained blockchain TPS, network finality latency, recovery time or hardware resource profile was measured. Compiler sample throughput is not blockchain throughput (M07). The benchmark plan below is proposed acceptance work; all metric cells intentionally remain unmeasured.

FIELD MANUAL / 09

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Table 11 Benchmark measurements to produce

Metric	Current measurement	Required experiment
Finalized unique TPS	NOT MEASURED	Count distinct successful transfers finalized in a stable interval; exclude retries, setup and query-only calls.
Latency p50/p95/p99	NOT MEASURED	Track signed submission → pool admission → inclusion → finality with monotonic timestamps.
Mempool capacity	NOT MEASURED	Sweep bounded queue occupancy, fees, invalid nonces and oversized input; measure drops and memory.
Validator scaling	NOT MEASURED	Repeat identical workload over 4, then larger authority sets with fixed hardware and latency.
State growth	NOT MEASURED	Track database size / finalized transaction and retained proof volume across a long soak.
RPC throughput	NOT MEASURED	Separate read-only requests, submission, subscriptions and abusive clients; include error rates.
Recovery time	NOT MEASURED	Crash and restore an isolated validator; compare finalized hash/root and catch-up time.
CPU/RAM/disk/network	NOT MEASURED	Record node, database, relayer and load-generator resources with sampled raw telemetry.

Known candidate bottlenecks are whole-asset finalization scans (H16), HashMap relayer event retention/rescan behavior (H10), blocking RPC work in async paths, serialized shared-state access, proof/receipt payload sizes and mandatory hook work that escapes transaction fee limits. These are static candidates, not measured hotspots. Rate-limiter unit tests do not establish whole-node DoS resistance.

Reproducible experimental protocol

Use four validator processes, separate non-signing load generators, disposable balances, fixed chain spec and exact binary/WASM/lockfile hashes. State CPU model, cores, RAM, disk model, kernel, network delay/loss profile and feature flags. Warm up, execute a fixed transaction mix, record a sustained interval and cool down; repeat enough runs to show variability. Report offered load, accepted load, finalized goodput and failures separately. Do not extrapolate compiler throughput or a single burst.

Failure-under-load scenarios: malformed/oversized input; two clients competing on a nonce; a lost relayer response; disk full in disposable storage; unavailable database; one validator killed; temporary partition; subscription churn; and reconnect during a runtime upgrade. Acceptance requires no supply drift or divergent finalized history. Restore/refund liveness must recover within an explicitly adopted operator SLO.

```
# Existing compiler-only measurement, not network TPS:
cargo run -p x3-bench -- --baseline
# Network/load harness must be implemented and reviewed first.
# Retain benchmark-results.csv using the schema in this artifact package.
```

OBSERVED RESULTS / MISSING ADVERSARIAL PROOF

10 Test and Verification Strategy

Current execution results are in evidence/verification-ledger.json. Failure is retained as evidence; tests that were blocked before execution are not counted as passed or failed test cases. New adversarial proof-router tests intentionally express the safe behavior and currently fail, demonstrating defects in unchanged source.

FIELD MANUAL / 10

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Figure 21 shows test evidence matrix: observed execution status, not coverage percentage.

Test evidence matrix

	Present	Ran	Passed
RPC algorithm	YES	YES	YES
Python parsing / types	YES	YES	YES
Python emitters	YES	YES	NO
Proof-router rejection	YES	YES	NO
Full workspace	YES	NO	NO
Two-node finality	YES	NO	NO
Contracts / migrations	YES	NO	NO

Build attempts ran; full workspace tests did not execute. This is not line coverage.

Figure 21 Test evidence matrix: observed execution status, not coverage percentage

Table 12 Executed test and build outcomes

Check	Observed outcome	Limit
Root check/test/clippy	All exit 101 (continued runs)	E0152 duplicate core in WASM build; no full workspace test result.
Release mainnet-rc1 / testnet check	Both exit 101	Same WASM build blocker.
Formatting	Exit 1	Formatting differences; logs retained.
RPC source harness	4 passed; 0 failed	Unchanged middleware module only.
Proof-router adversarial harness	0 passed; 3 failed	Invalid proofs accepted with production feature.
Selected Python DSL tests	6 passed; 4 failed	Emitter ProofRequiredError; not a live trade test.
npm / pnpm / Python alias commands	Unavailable tools/dependencies	pnpm/python absent on PATH; isolated JS installation omitted node_modules.
Contracts	forge/anchor not found	No contract build/test/deploy.
Python migration collection	1 collection error	alembic missing; no DB mutation.
Configured / unsuppressed audit	0 / 53 vulnerability matches	Cached DB freshness unknown; runtime reachability not established.
Fake-code scan	54,017 matching locations; grep exit 2	Traversal/read errors; candidate matches include tests/vendor/docs, not all defects.

Unit tests

What could break: Exact function boundaries and arithmetic. Fixtures/environment and procedure: Small deterministic inputs plus error cases; reject malformed lengths, overflow and unauthorized origin.

Pass/fail: All new regression cases pass; retain named test log. CI placement: Per PR. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
cargo test -p <affected-package>
```

Property tests

What could break: Supply/fee/escrow conservation. Fixtures/environment and procedure: Generate transfer/lock/claim/refund sequences over bounded assets, nonces and limits.

Pass/fail: Invariant holds after each successful or failed operation; minimize failing seeds. CI placement: Per PR + nightly. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Proposed property harness in runtime/pallet test modules

Fuzz tests

What could break: Parser/VM/proof malformed input. Fixtures/environment and procedure: Corpus of RLP/SCALE/bytecode/header boundary cases; no production credentials.

Pass/fail: No crash, uncontrolled allocation or acceptance of invalid proof. CI placement: Nightly. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Use inventoried pallet fuzz manifests after dependency/build repair

Integration tests

What could break: RPC to real runtime state. Fixtures/environment and procedure: Fresh node and ephemeral keys; signed transaction through network RPC.

Pass/fail: Inclusion/finality receipt and cross-node persisted state agree. CI placement: Required gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
cargo test -p x3-chain-node --lib
```

Migration tests

What could break: Old schema to current state. Fixtures/environment and procedure: Disposable database/runtime snapshots for every supported prior version.

Pass/fail: No loss/duplication; pre/post invariants and replayed roots agree. CI placement: Required upgrade gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Runtime try-runtime path + SQL/Alembic upgrade tests

Multi-node tests

What could break: Consensus and finality. Fixtures/environment and procedure: 4 distinct validator processes and keys with isolated ports/data.

Pass/fail: All honest nodes finalize same hash/root; no ignored critical case. CI placement: Required gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
Repair node service tests and Zombienet config
```

Partition tests

What could break: Split and reconnect. Fixtures/environment and procedure: Introduce bounded network partitions without external peers.

Pass/fail: No conflicting finality; majority recovery and catch-up demonstrated. CI placement: Nightly/release. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
Proposed network-fault harness
```

Consensus adversary tests

What could break: Equivocation, outsider votes. Fixtures/environment and procedure: Valid authority fixture and distinct unauthorized keys.

Pass/fail: Unauthorized/duplicate/wrong-set evidence rejected; honest liveness preserved. CI placement: Required gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
cargo test -p x3-chain-node --lib
```

Contract tests

What could break: HTLC and verifier semantics. Fixtures/environment and procedure: Foundry fuzz/invariant tests; Anchor local validator with funded test accounts.

Pass/fail: Invalid proofs never release; claim/refund once only; accounting conserved. CI placement: Required contract gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
forge test --root X3-contracts/evm; Anchor workspace local tests
```

Cross-chain tests

What could break: Authenticated deposit/release. Fixtures/environment and procedure: Local chains or recorded independent proofs; relayer isolated from live keys.

Pass/fail: Exact asset/sender/recipient/amount and finality verified at each leg. CI placement: Required bridge gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

```
Audit proof harness + real gateway runtime tests
```


Load / soak tests

What could break: Sustained resource pressure. Fixtures/environment and procedure: Fixed workload; resource telemetry and unique finalized transaction IDs.

Pass/fail: No unbounded growth/supply drift; adopted latency/error SLO met. CI placement: Release/nightly. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Proposed load harness; schema in benchmark-results.csv

Chaos tests

What could break: Disk/DB/process failure. Fixtures/environment and procedure: Disposable services, controllable storage quotas and process supervision.

Pass/fail: No corrupted accepted state; bounded recovery and correct retries. CI placement: Release. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Proposed isolated chaos harness

Upgrade tests

What could break: Runtime and authority transition. Fixtures/environment and procedure: Old and new binaries with pinned state snapshots and session schedule.

Pass/fail: Migration invariants and one finalized history; compatible rollback defined. CI placement: Required upgrade gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

cargo test -p x3-chain-runtime --features try-runtime

Backup / restore tests

What could break: Data reconstruction. Fixtures/environment and procedure: Authenticated snapshot plus independent finalized root and off-host copy.

Pass/fail: Tamper rejected; restored node reaches recorded root and catches up. CI placement: Release + periodic. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Review then exercise scripts/snapshot-restore.sh on disposable paths

Security tests

What could break: Critical regression suite. Fixtures/environment and procedure: The C01-C03/H01-H18 acceptance cases and source-bound evidence.

Pass/fail: Zero open applicable Critical/High; no waived fund-safety tests. CI placement: Required gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Affected package tests and independent review retest

End-to-end tests

What could break: Wallet to finalized settlement. Fixtures/environment and procedure: Real SDK/wallet encoding, pool, node, receipt, indexing and restart.

Pass/fail: User-visible success only after required finality; failures/refunds displayed honestly. CI placement: Required gate. Evidence: logs, source/feature hashes, fixture digests and relevant state roots in a per-run immutable directory.

Implement scoped e2e flow in tests/e2e

FRESH MACHINE / VALIDATOR / RECOVERY

11 Operations and Deployment

A new operator cannot be told that this source state safely launches a public validator: audited build commands fail, no signed release/genesis was verified, and critical runtime defects remain. Installation and local launch commands below are inferred from the repository and are blocked procedures, not instructions to activate a live network now.

FIELD MANUAL / 11

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

```
# In a fresh isolated checkout matching provenance + dirty-source hashes:
rustc --version
cargo --version
protoc --version
clang --version
cargo check --workspace
cargo test --workspace
cargo clippy --workspace --all-targets -- -D warnings
cargo build --locked --release -p x3-chain-node --features mainnet-rc1
# BLOCKED by H18 in this audit. After repair, local-only smoke:
./target/release/x3-chain-node --chain dev --tmp --rpc-port 9933 --validator --alice --no-telemetry
```

Repository Rust pin is 1.90.0 with rust-src/clippy/rustfmt and wasm32-unknown-unknown. The build also needs native compiler/protobuf/LLVM tooling and cached or approved locked dependencies. Root package.json specifies pnpm 10.15.1; no pnpm installation was available on audit PATH. The JS copy deliberately excluded node_modules, so missing tsc/vitest is an isolated-install blocker, not proof that the original installed applications cannot build. Python 3.10 with pytest was usable as /usr/bin/python3; python alias was absent. Foundry/Anchor were unavailable. Do not download and execute installer scripts without reviewing/pinning them.

Production deployment checklist

Require source-bound signed artifacts; verify chain/genesis/WASM identity; use least-privilege user; isolate DB/metrics/admin ports; configure TLS RPC and rate limits; verify dependency readiness; keep documented rollback and on-call ownership. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

Validator launch checklist

Generate separate session/custody keys; validate authority public-key uniqueness and balances; confirm correct spec and distinct data path; expose only P2P; verify peers, authoring and finalized head; prove restart without key leakage. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

Public testnet launch checklist

Close P0s; run 4-node fault tests; publish exact scope and disabled routes; establish telemetry/alerts, abuse controls, faucet limits, incident channel and restore proof; require accountable sign-off. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

Mainnet launch checklist

Zero applicable Critical/High; independent retest; reproducible release/WASM/genesis; custody rotation/recovery; supply/fee audit; validated upgrade/rollback; nonwaivable safety gates; approve exact hashes. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

Emergency shutdown checklist

Pause affected route/module through documented authorized control; avoid halting refunds unnecessarily; stop compromised signers; preserve logs/state evidence; coordinate validators; verify balances and finalized history before resuming. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

Upgrade and rollback checklist

Replay old snapshot into candidate; run migrations and invariant checks; verify session boundary; canary under identical spec; retain prior binary/state; never blindly run an old binary over a changed schema. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

Backup restoration drill checklist

Stop only the disposable validator; hash/authenticate archive and manifest; restore to empty isolated path; verify genesis, schema and recorded finalized root; reconnect to isolated peers and prove catch-up; record RPO/RTO measurements. Status: NOT PROVEN for this audited state; checklist text earns no readiness credit.

packaging/systemd/x3-validator.service uses an unprivileged user and filesystem/process hardening, providing a useful starting point. Inspect actual effective service configuration before use. Some infra assets are development scaffolding (including dev-mode secret service and unpinned image tags); these must not be promoted wholesale to production. No infrastructure secrets were copied into this package.

DEPENDENCIES / OWNERS / ACCEPTANCE

12 The Completion Blueprint

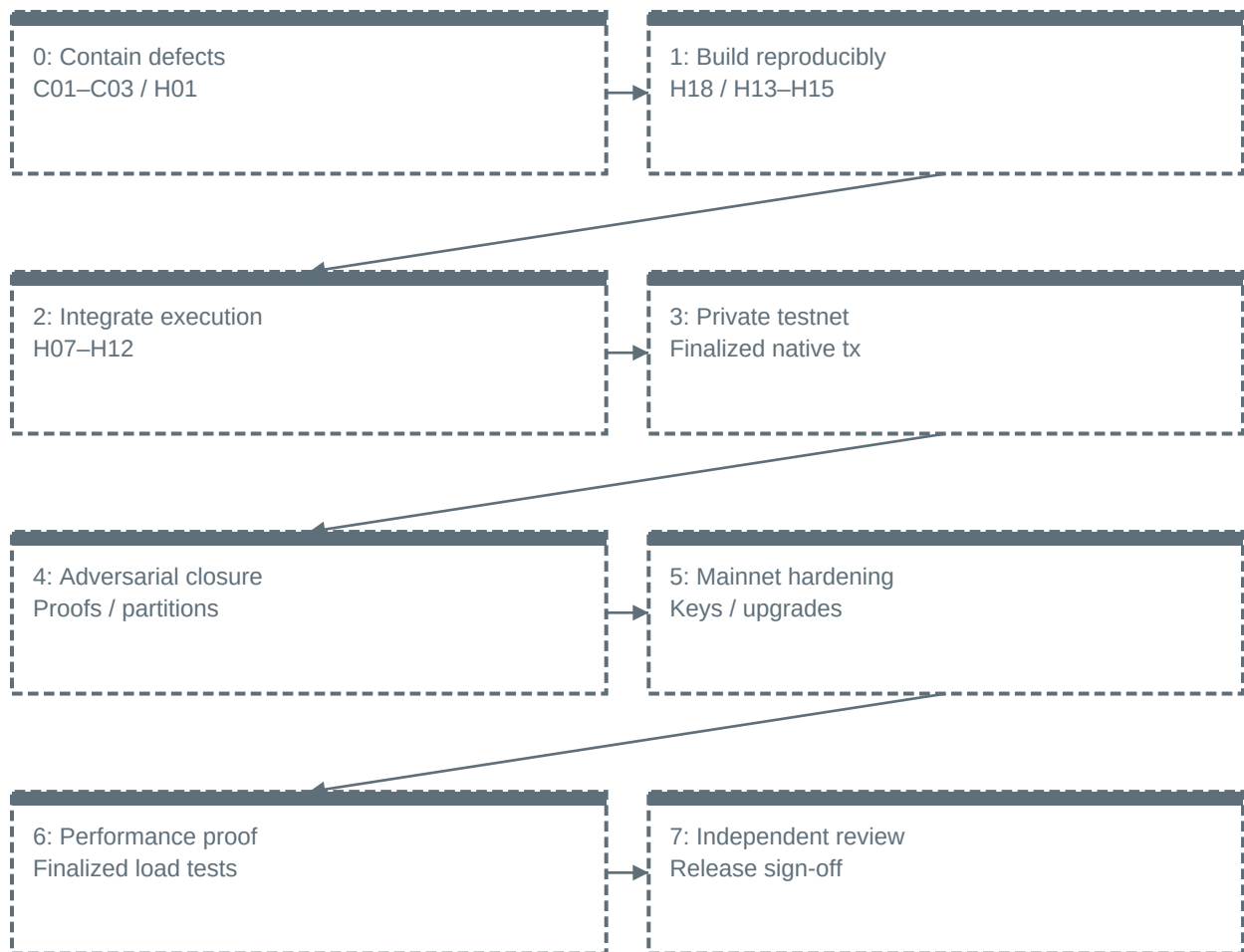
Each confirmed finding becomes an engineering work item with a unique ID, exact affected files, dependencies, acceptance criteria, test command, owner and rollback plan. The task register is machine-readable in recovery-plan.json. Ordinal S/M/L/XL estimates communicate relative complexity only. Team capacity and unknown protocol decisions make calendar commitments unjustified.

FIELD MANUAL / 12

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Figure 12 shows critical-path completion roadmap.

Critical-path completion roadmap

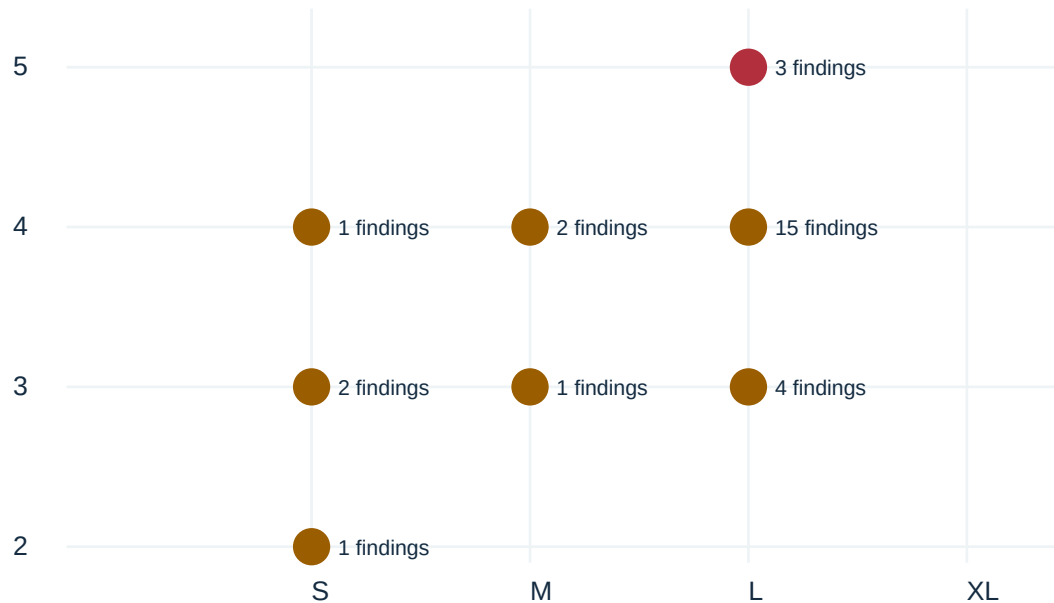


I implemented/unverified P partial D disconnected X unsafe V verified N proposed
Proposed sequence. No milestone is represented as already passed.

Figure 12 Critical-path completion roadmap

Figure 23 shows effort-versus-impact prioritization: ordinal estimates and grouped finding counts.

Effort versus impact — ordinal prioritization

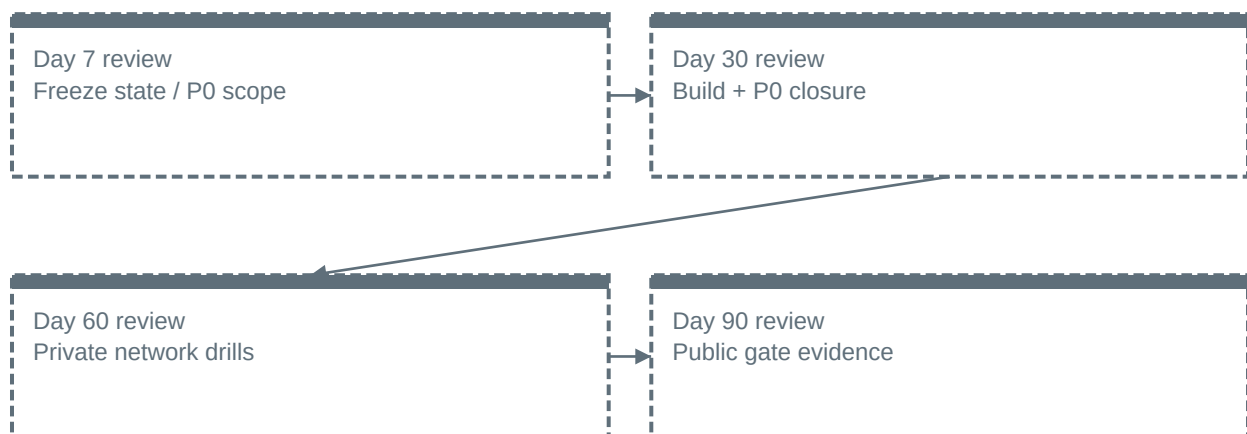


Complexity S/M/L/XL; impact 2–5. Group counts; no person-day or cost estimates.

Figure 23 Effort-versus-impact prioritization: ordinal estimates and grouped finding counts

Figure 13 shows 7 / 30 / 60 / 90-day planning horizons.

7 / 30 / 60 / 90-day planning horizons



I implemented/unverified P partial D disconnected X unsafe V verified N proposed

Review horizons only; dependent on staffing and acceptance gates, not delivery promises.

Figure 13 7 / 30 / 60 / 90-day planning horizons

Table 13 Engineering phases

Phase / purpose	Work items	Exit condition
0 Stop-the-line security and integrity	C01,C02,C03,H01,H03,H04,H06	Contain unsigned/structural trust paths and validate their replacements.
1 Restore core build and proof controls	H18,H13,H14,H15	Produce a clean build and gates that cannot hide failed evidence.
2 Complete runtime integration	H02,H05,H07,H08,H09,H10,H11,H12,H17,M05	Wire persistent execution and authenticated submission; no path-existence credit.
3 Public testnet prerequisites	M02,L01	Require isolated multi-node finality/recovery and reproducible operator setup.
4 Adversarial stabilization	All safety finding retests	Run partitions, malformed proofs, replay, rollback and unauthorized-origin tests.
5 Mainnet hardening	H16,M03,M04,M06	Bound resources; prove event delivery, key lifecycle and restore/upgrade.
6 Performance optimization	M07	Measure genuine finalized workloads before optimization.
7 Independent review and approval	M01 + complete gate register	Retest closure independently and approve exact release/spec hashes.

Parallelization potential: SDK repairs, gateway startup, CI corrections and independent proof-verifier implementations can proceed once interfaces and source state are fixed. Final integration must serialize around runtime metadata, proof envelope format, genesis and storage migrations. Do not run multiple independent writers against custody state or shared migration databases.

FIX-C01 / External header validation accepts unproved claims

P0 | Protocol security | Complexity L | Risk Critical. Dependencies: None specific; H18 controls runtime execution proof.

Files: pallets/cross-chain-validator/src/lib.rs:181; pallets/cross-chain-validator/src/lib.rs:332; pallets/cross-chain-validator/src/lib.rs:279; runtime/src/lib.rs:2412; runtime/src/lib.rs:980

Implementation: Use authenticated chain-specific finalized-header clients with trusted validator/checkpoint transitions, verified signatures, bounded proofs, and authenticated parent links. Reject unsupported chains.

Acceptance: Arbitrary hashes, duplicate signers, self-selected validator sets, future heights and invalid membership proofs are rejected without writes; canonical external fixtures pass.

```
cargo test -p pallet-cross-chain-validator --features std
```

Retain: closure/C01/manifest.json with exact source/features, raw test logs and relevant roots/receipts. Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-C02 / Unsigned finality anchors make certificate checks circular

P0 | Protocol security | Complexity L | Risk Critical. Dependencies: None specific; H18 controls runtime execution proof.

Files: pallets/x3-atomic-kernel/src/lib.rs:884; pallets/x3-atomic-kernel/src/lib.rs:1118;
pallets/x3-atomic-kernel/src/lib.rs:1416

Implementation: Validate an actual GRANDPA justification or authenticated authority certificate bound to block hash, number, set ID and chain ID; remove arbitrary unsigned anchor admission.

Acceptance: Forged and conflicting anchors cannot enter pool or dispatch; independently verified justification anchors the exact finalized block; late legitimate anchors remain recoverable.

```
cargo test -p pallet-x3-atomic-kernel --features std
```

Retain: closure/C02/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-C03 / Unsigned rollback receipts trust attacker-selected prior state

P0 | Protocol security | Complexity L | Risk Critical. Dependencies: FIX-C02

Files: pallets/x3-atomic-kernel/src/lib.rs:1030; pallets/x3-atomic-kernel/src/lib.rs:1150;
pallets/x3-atomic-kernel/src/lib.rs:1289; pallets/x3-atomic-kernel/src/vm_revert.rs:342;
runtime/src/lib.rs:2675

Implementation: Generate diffs inside transactional execution, bind them to executor/leg/pre-state/access set, and validate both old and current values. Roll back in reverse dependency order with complete failure propagation.

Acceptance: Unrelated caller/diff and stale pre-state are rejected; injected rollback failure leaves no partial state; reverse-order replay restores exact pre-root.

```
cargo test -p pallet-x3-atomic-kernel --features std
```

Retain: closure/C03/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H01 / Production proof-router alternatives accept arbitrary bytes

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-C01

Files: crates/x3-verification-router/src/lib.rs:323; crates/x3-verification-router/src/lib.rs:355;
crates/x3-verification-router/src/lib.rs:276; pallets/x3-crosschain-gateway/src/lib.rs:1162;
runtime/src/lib.rs:2479

Implementation: Remove structural verifiers from production and implement signature membership, unique quorum, chain binding and event inclusion; retain governance admission only as defense in depth.

Acceptance: All three isolated rejection tests pass; actual signed finalized fixtures pass through the runtime gateway, not only the helper.

```
cargo test --manifest-path audit-harness/proof/Cargo.toml
```

Retain: closure/H01/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H02 / EVM receipt verifier lacks trusted roots and misbinds inclusion

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-C01

Files: crates/x3-verification-router/src/evm_receipt.rs:737;
crates/x3-verification-router/src/evm_receipt.rs:593;
crates/x3-verification-router/src/evm_receipt.rs:335

Implementation: Use canonical typed receipts and RLP(index), verify the exact receipt value under an authenticated receiptsRoot, bind gateway/event/amount/sender/recipient/chain, and verify finality separately.

Acceptance: Cross-check independent execution-client proof vectors, tampered receipt values and false heads; require both authentic root and exact inclusion.

```
cargo test -p x3-verification-router --features production
```

Retain: closure/H02/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H03 / Bitcoin verification uses asserted confirmations and incomplete vault approval

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-H01

Files: crates/x3-verification-router/src/lib.rs:557; crates/x3-verification-router/src/lib.rs:617;
crates/x3-bitcoin-vault/src/lib.rs:324; crates/x3-bitcoin-vault/src/lib.rs:450

Implementation: Implement persistent best-work header validation with checkpoints/difficulty rules and transaction output proofs; verify distinct authorized signatures over exact deposit/withdrawal payloads.

Acceptance: Reject invented tips, wrong outputs, wrong recipient/value, repeated approvals and forged signatures; restart and reorg tests preserve accounting.

```
cargo test -p x3-bitcoin-vault && cargo test -p x3-verification-router --features production
```

Retain: closure/H03/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H04 / Public misbehavior report has no evidence and does not slash currency

P0 | Protocol security | Complexity L | Risk High. Dependencies: None specific; H18 controls runtime execution proof.

Files: pallets/x3-consensus/src/lib.rs:263; pallets/x3-consensus/src/lib.rs:381; pallets/x3-consensus/src/tests/slashing.rs:31; runtime/src/lib.rs:1063

Implementation: Route offenses through validated equivocation/unavailability evidence with unique offence IDs; reserve real stake, slash it atomically and coordinate session removal.

Acceptance: Invalid/replayed evidence cannot slash; total reserved balances and treasury/burn conservation reconcile after each valid offence; removed keys cannot author/vote.

```
cargo test -p pallet-x3-consensus --features std
```

Retain: closure/H04/manifest.json with exact source/features, raw test logs and relevant roots/receipts. Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H05 / Validator rotation ignores requested activation delay at session boundary

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-H04

Files: pallets/x3-consensus/src/lib.rs:229; pallets/x3-consensus/src/lib.rs:412; runtime/src/lib.rs:1042

Implementation: Use one activation state machine for queued keys, session selection and metadata. Reject empty/duplicate/unkeyed sets and enforce the same scheduled boundary.

Acceptance: A change delayed over several sessions is not activated early; restart and session transition preserve identical authority sets on all nodes.

```
cargo test -p pallet-x3-consensus --features std
```

Retain: closure/H05/manifest.json with exact source/features, raw test logs and relevant roots/receipts. Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H06 / Flash-finality opt-in disables GRANDPA without a proven replacement

P0 | Protocol security | Complexity L | Risk High. Dependencies: None specific; H18 controls runtime execution proof.

Files: node/src/service.rs:656; node/src/service.rs:962; node/src/service.rs:1937; crates/flash-finality/src/lib.rs:389

Implementation: Keep GRANDPA mandatory until a separately audited finality engine proves authority membership, safety and liveness; reject incompatible flags or missing keys at startup.

Acceptance: Wrong keys/sets cannot form certificates; missing keys fail startup; partition, equivocation and recovery tests establish one finalized history.

```
cargo test -p x3-chain-node --lib
```

Retain: closure/H06/manifest.json with exact source/features, raw test logs and relevant roots/receipts. Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H07 / Default mini-EVM executes against disposable synthetic state

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-C03

Files: crates/evm-integration/src/mini_evm.rs:102; crates/evm-integration/src/mini_evm.rs:148; pallets/x3-kernel/src/wasm_adapters.rs:34; runtime/src/lib.rs:1564

Implementation: Connect actual runtime state, caller/value, logs, state commitment and gas charging through a transactional interpreter; run canonical EVM conformance tests.

Acceptance: A contract SSTORE persists across blocks/restart, a second user cannot impersonate the first, out-of-gas fully reverts and fees reconcile.

```
cargo test -p x3-chain-runtime --features frontier
```

Retain: closure/H07/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H08 / Ethereum submission RPC bypasses the transaction-pool lifecycle

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-H07

Files: node/src/rpc_frontier.rs:457; runtime/src/lib.rs:3425; node/src/rpc.rs:341

Implementation: Decode and verify canonical signed transactions and submit an appropriate runtime extrinsic through the pool; report receipts only after inclusion and finality.

Acceptance: Submit a signed transfer over RPC, observe pool admission, block inclusion, finality and persisted balance change on another node; reject replay/wrong chain ID.

```
cargo test -p x3-chain-node --features frontier --lib
```

Retain: closure/H08/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H09 / Relay manual extrinsic encoding and signing do not match runtime

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-H01, FIX-H08

Files: crates/x3-relayer/src/submitter.rs:437; crates/x3-relayer/src/submitter.rs:568; runtime/src/lib.rs:850

Implementation: Use metadata-derived typed calls and the exact signed extensions, chain/genesis versions and custody signer; introduce an explicit governed relayer authorization model.

Acceptance: Exact production runtime decodes, authenticates, includes and finalizes generated deposit/release transactions; no fallback signer or zero-chain context.

```
cargo test -p x3-relayer
```

Retain: closure/H09/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H10 / Relay checkpoints can skip failed events and are not durable

P0 | Protocol security | Complexity L | Risk High. Dependencies: FIX-H09

Files: crates/x3-relayer/src/main.rs:535; crates/x3-relayer/src/main.rs:795;
crates/x3-relayer/src/main.rs:827

Implementation: Persist transactional inbox/outbox and per-event delivery state; checkpoint only a contiguous completed finalized range with reorg rollback and idempotent retries.

Acceptance: Fail submission for an early event, process later events, restart and recover the failed event exactly once; bound RPC scans and memory.

```
cargo test -p x3-relayer
```

Retain: closure/H10/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H11 / Gateway binary never starts its REST/GraphQL implementation

P1 | Backend / SRE | Complexity L | Risk High. Dependencies: None specific; H18 controls runtime execution proof.

Files: crates/x3-gateway/src/main.rs:1; crates/x3-gateway/Cargo.toml:12;
crates/x3-gateway/src/rest.rs:85

Implementation: Implement validated configuration loading, database migrations, cache setup, router binding and graceful shutdown; expose dependency readiness separately from liveness.

Acceptance: Start the built binary against disposable services, serve actual API/GraphQL data, fail readiness when DB is down, and recover after restart.

```
cargo test -p x3-gateway && cargo run -p x3-gateway
```

Retain: closure/H11/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H12 / SDK EVM and SVM encoding contains production placeholders

P1 | SDK engineer | Complexity L | Risk High. Dependencies: None specific; H18 controls runtime execution proof.

Files: packages/ts-sdk/src/evm.ts:140; packages/ts-sdk/src/evm.ts:154;
packages/ts-sdk/src/evm.ts:440; packages/ts-sdk/src/svm.ts:230

Implementation: Use canonical Keccak address/selector derivation and canonical Solana message compilation with real account metas and program indices.

Acceptance: Match independent reference vectors for EVM selectors/addresses and multi-account SVM messages, then finalize one transaction per advertised protocol.

```
npm --prefix packages/ts-sdk test -- --runInBand
```

Retain: closure/H12/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H13 / CI make recipes swallow failures and reference missing targets

P0 | Release engineering | Complexity S | Risk High. Dependencies: None specific; H18 controls runtime execution proof.

Files: Makefile:19; Makefile:29; Makefile:38; .github/workflows/production-gate.yml:25

Implementation: Preserve exit codes, use exact package names and define/reconcile workflow targets; test gates against a deliberately failing test in disposable fixtures.

Acceptance: Cargo failure produces nonzero make/CI exit; every workflow command resolves; mainnet-required aggregate cannot pass with skipped/missing jobs.

```
make test-x3-wallet; make -n test-node-build
```

Retain: closure/H13/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H14 / Release gate accepts stale build artifacts without source binding

P0 | Release engineering | Complexity M | Risk High. Dependencies: FIX-H13

Files: scripts/mainnet_release_gate.py:84; scripts/mainnet_release_gate.py:118

Implementation: Build from a clean locked commit, verify compiler/runtime/features, regenerate genesis, sign a source-bound manifest and reject dirty or stale evidence.

Acceptance: Mutating source or substituting a binary/spec invalidates evidence and blocks release; two builders reproduce the same WASM.

```
python3 scripts/mainnet_release_gate.py
```

Retain: closure/H14/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H15 / Dependency audit pass masks unresolved advisory matches

P0 | Security / dependency engineering | Complexity L | Risk High. Dependencies: FIX-H13

Files: .cargo/audit.toml:1; Cargo.lock:1; evidence/advisories-unfiltered.json

Implementation: Triage each advisory against shipped binaries/features; upgrade affected paths and narrowly justify time-limited exceptions with technical evidence. Verify fresh official advisory data before release.

Acceptance: No reachable unmitigated critical/high advisories; exceptions have owners, expiry, exact dependency paths and regression evidence.

```
cargo audit --no-fetch --stale --no-yanked --file Cargo.lock --json
```

Retain: closure/H15/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H16 / Supply invariant finalization work is unbounded by asset count

P0 | Runtime performance | Complexity L | Risk High. Dependencies: FIX-C01

Files: pallets/x3-supply-ledger/src/lib.rs:158; pallets/x3-supply-ledger/src/lib.rs:166

Implementation: Bound asset cardinality/work or use an incremental authenticated aggregate; benchmark and reserve hook weight with proof size accounting.

Acceptance: Worst-case asset count fits declared block budget; adversarial asset creation cannot make next-block mandatory work unbounded.

```
cargo test -p pallet-x3-supply-ledger --features std
```

Retain: closure/H16/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H17 / EVM external verifier is blocked in production and bypassable by mode

P0 | Smart-contract security | Complexity L | Risk High. Dependencies: FIX-H01

Files: X3-contracts/evm/contracts/EvmReceiptVerifier.sol:205;

X3-contracts/evm/contracts/EvmReceiptVerifier.sol:183;

X3-contracts/evm/contracts/EvmReceiptVerifier.sol:73

Implementation: Deploy a real supported verifier with domain/set binding and unique quorum; remove bypass from funded deployments and govern rotations with delays.

Acceptance: Real proof accepted, zero/duplicate/forged signatures rejected; funded deployment cannot enable structural bypass; owner rotation preserves trust policy.

```
forge test --root X3-contracts/evm
```


Retain: closure/H17/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M01 / Readiness documentation contradicts executable evidence

P2 | Release / technical writing | Complexity S | Risk Medium. Dependencies: None specific; H18 controls runtime execution proof.

Files: CURRENT_MAINNET_STATUS.md:3; CURRENT_MAINNET_STATUS.md:63;
FEATURE_REGISTRY.toml:18; evidence/consistency.log

Implementation: Generate status from commit-bound feature proof criteria and preserve blocked/failed results. Remove operational claims that only have path-existence evidence.

Acceptance: Any failed core proof lowers its status; stale commit evidence is rejected; readiness prose and matrix derive from the same data.

```
bash scripts/check-readiness-consistency.sh
```

Retain: closure/M01/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M02 / Finality and multi-node proof tests are ignored

P1 | Distributed-systems QA | Complexity L | Risk Medium. Dependencies: FIX-H13

Files: node/src/service.rs:2429; node/src/service.rs:2580; node/src/service.rs:2664

Implementation: Run each node in a separate process with isolated logger/ports/data; hard-gate finalized transaction and reconnect recovery tests.

Acceptance: At least four validators agree on finalized hash/root through restart and one-node failure; ignored critical tests count as release failure.

```
cargo test -p x3-chain-node two_validator_nodes_submit_on_first_observe_finalized_bridge_state_on_second  
-- --ignored --nocapture
```

Retain: closure/M02/manifest.json with exact source/features, raw test logs and relevant roots/receipts.
Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M03 / Gateway health and event spines do not prove dependencies

P2 | Observability / SRE | Complexity L | Risk Medium. Dependencies: FIX-H11

Files: crates/x3-gateway/src/rest.rs:248; runtime/src/lib.rs:11; runtime/src/lib.rs:24

Implementation: Separate liveness/readiness, probe DB/indexer lag and durable event delivery, and make missing required consumers explicit startup/readiness failures.

Acceptance: Stop database/event sink and see readiness fail; replay missed events exactly once after recovery.

```
cargo test -p x3-gateway
```

Retain: closure/M03/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M04 / Restore validates archive shape, not trusted checksum or chain state

P2 | SRE / storage | Complexity L | Risk Medium. Dependencies: FIX-H14

Files: scripts/snapshot-restore.sh:130; scripts/snapshot-restore.sh:141

Implementation: Verify authenticated manifest, confined archive paths, chain/genesis/schema identity and finalized state before activation.

Acceptance: Tampered archive fails before extraction; restored node catches up and matches independently recorded finalized state.

```
bash -n scripts/snapshot-restore.sh
```

Retain: closure/M04/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M05 / Python DSL emitter tests fail under the current proof contract

P1 | Language / SDK | Complexity L | Risk Medium. Dependencies: FIX-H01

Files: x3-lang/emitter/x3.py:60; x3-lang/tests/test_emitter.py:1; evidence/dsl-tests.log

Implementation: Supply real verification fixtures and reconcile emitters, runner and CLI contracts without weakening fail-closed proof checks.

Acceptance: All emitter tests pass with independently validated proof fixtures and reject empty/tampered bundles; one local end-to-end emission is consumed by the real runtime.

```
/usr/bin/python3 -m pytest -q x3-lang/tests/test_emitter.py
```

Retain: closure/M05/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M06 / Desktop defaults select an unregistered chain

P2 | Desktop / wallet | Complexity S | Risk Medium. Dependencies: None specific; H18 controls runtime execution proof.

Files: apps/x3-desktop/src/blockchain/ChainManager.ts:59;
apps/x3-desktop/src/blockchain/ChainManager.ts:117

Implementation: Register configured enabled adapters and select an initialized network by verified chain/genesis identity; use secure transport for public endpoints.

Acceptance: Default startup obtains the intended adapter, missing network fails clearly, and wrong genesis cannot be signed against.

```
npm --prefix apps/x3-desktop test
```

Retain: closure/M06/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-M07 / Optimizer throughput is not network TPS evidence

P3 | Performance engineering | Complexity M | Risk Medium. Dependencies: None specific; H18 controls runtime execution proof.

Files: crates/x3-bench/src/main.rs:3; crates/x3-bench/src/runner.rs:251

Implementation: Keep compiler metrics labeled; add finalized unique transaction load tests with latency histograms and resource/recovery evidence.

Acceptance: Publish reproducible sustained finalized TPS, rejection rate, p50/p95/p99 finality latency and hardware/profile/commit details; no synthetic metric substitution.

```
cargo run -p x3-bench -- --baseline
```

Retain: closure/M07/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-L01 / Fresh-machine helper does not verify the required environment

P2 | Developer experience | Complexity S | Risk Low. Dependencies: None specific; H18 controls runtime execution proof.

Files: scripts/fresh_machine_check.sh:5; Makefile:61; rust-toolchain.toml:3

Implementation: Validate exact tools/versions, native libraries and lockfile installs; distinguish unavailable tool from failed project test.

Acceptance: Clean image builds all intended targets and runs real smoke tests with explicit offline/cache prerequisites.

```
bash scripts/fresh_machine_check.sh
```

Retain: closure/L01/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

FIX-H18 / Workspace verification fails in WASM runtime build

P1 | Rust build / release engineering | Complexity M | Risk High. Dependencies: None specific; H18 controls runtime execution proof.

Files: runtime/build.rs:1; rust-toolchain.toml:3; evidence/check-complete.log; evidence/test-complete.log; evidence/clippy-complete.log

Implementation: Reconcile pinned rust-src/target/build-std configuration and dependency build graph in a fresh build directory; preserve both native and WASM validation without skip-WASM workarounds.

Acceptance: From the frozen source state, all required workspace checks and release/testnet feature builds exit zero and produce source-bound artifacts.

```
cargo check --workspace && cargo test --workspace && cargo clippy --workspace --all-targets -- -D warnings
```

Retain: closure/H18/manifest.json with exact source/features, raw test logs and relevant roots/receipts.

Rollback: Retain the previous signed binary and state snapshot; never downgrade across changed storage without a tested reverse migration. Keep affected entrypoints disabled if rollback cannot preserve state.

Kanban and risk retirement

Current board: all findings are Open; no implementation repairs were authorized by the read-only audit. Ready-for-work means scope, owner and acceptance are agreed. In-progress means code/tests exist in an isolated branch. Review-ready requires current passing evidence. Closed requires independent retest and integrated runtime proof. Risk burn-down is the count of independently closed findings over time; no invented falling curve is plotted.

Testnet definition of done: safe disabled scope, clean build, four-validator finalized transaction proof, replay/rollback/supply closure, network/restart/recovery tests and operational readiness. Mainnet adds independent audit closure, authenticated enabled external routes, custody/genesis ceremony, upgrade/disaster-recovery drills, measured capacity and exact artifact sign-off. A calendar date cannot substitute for either definition.

HONEST MILESTONES / TECHNICAL PROOF

13 Funding and Partnership Readiness

The technically compelling opportunity is a unified native execution and asset-accounting platform with explicit cross-VM intent/settlement abstractions. The investable engineering case is a milestone program that closes verified gaps, not a claim of a finished high-throughput trustless network. No market size, valuation, partnership or endorsement is asserted.

FIELD MANUAL / 13

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Table 14 Milestone-based funding packages

Milestone / sponsor value	Deliverable and technical proof	Resources / success / risk
Reproducible foundation	Repair H18/H13/H14; clean locked builds, signed manifests and failure-preserving gates.	Rust/release engineer + CI runners; every required command passes; dependency compatibility risk.
Authenticated proof boundary	Close C01/C02/H01-H03/H17 with independent canonical proof fixtures and rejection tests.	Protocol/crypto + contract security expertise; zero forged proof acceptance; high protocol complexity.
Atomic execution integrity	Close C03/H07/H08; persisted VM state and exact rollback under injected failure.	Runtime/VM engineer + adversarial QA; root/balance conservation across failure; integration risk.
Private validator network	Four-node signed transfer/finality/restart/partition demonstration with raw evidence.	Isolated hosts/processes, monitoring and storage; no divergent finalized roots; operational uncertainty.
Usable developer path	Canonical SDK encoders, real gateway startup, repaired emitter contract and honest receipts.	SDK/backend/language engineers; one real end-to-end user flow; metadata/version alignment risk.
Launch assurance	Independent review, custody/upgrade/restore drills and public gate decision for exact release.	External auditors, operator time, testnet infrastructure and incentive budget determined after scope; no unresolved nonwaivable safety issue.

Sponsors should request exact commit plus dirty-state provenance, independent build reproduction, test logs with skipped counts, canonical proof vectors, finalized transaction receipts from multiple nodes, resource measurements and independent security retest. Hardware needs cannot be honestly priced without a sustained-load baseline. Budget testnet incentives only after abuse controls and fund-safety gates are demonstrably effective.

Claims to avoid today: mainnet-ready, 100% complete, trustless external bridges, proven public-network TPS, audited cryptographic security, fully Ethereum/Solana-compatible execution, or a working gateway inferred solely from dependencies.

HARD THRESHOLDS / RESPONSIBILITY / FAILURE

14 Final Launch Gates

These are proposed objective release requirements, not retroactive evidence that a gate passed. All six stages are currently unproven or blocked for the audited state. High-risk safety gates are nonwaivable. Hosted branch protection, deployed ownership and operator identities were not available to verify.

FIELD MANUAL / 14

Base commit 6a24d8cf38f2522ddf9ae0b47011fd59a9984208
Audited 2026-09-05 with pre-existing working-tree changes.

Figure 22 shows launch gate dashboard.

Launch gate dashboard

Internal devnet		0
Private multi-node		0
Public testnet		0
Incentivized testnet		0
Release candidate		0
Mainnet		0

0 = gate not established. Requirements are in Chapter 14; no waiver of safety gates.

Figure 22 Launch gate dashboard

Internal devnet

Requirement: Current locked node/runtime build, correct local genesis and basic signed transfer.

Measurement / pass threshold: All required local build/format checks pass; one transfer included and persisted after restart. Responsible role: Build/release engineer. Evidence artifact: build logs, binary/WASM hashes, genesis hash, receipt and restart state.

Failure response: stop promotion, retain evidence, contain the affected path, repair and independently re-run the failing acceptance cases. Waiver policy: no waiver for fund safety, consensus, key management or state integrity; non-safety deviations require a signed scope reduction with owner and expiry, never a falsified pass. Current status: BLOCKED / NOT ESTABLISHED.

Private multi-node testnet

Requirement: Four distinct validators; replay, partition and crash tests. Measurement / pass threshold: One finalized hash/root across honest nodes; no double debit; recovery succeeds; no ignored critical test. Responsible role: Consensus lead. Evidence artifact: topology/config hashes, per-node logs, finalized roots, fault timeline.

Failure response: stop promotion, retain evidence, contain the affected path, repair and independently re-run the failing acceptance cases. Waiver policy: no waiver for fund safety, consensus, key management or state integrity; non-safety deviations require a signed scope reduction with owner and expiry, never a falsified pass. Current status: BLOCKED / NOT ESTABLISHED.

Public testnet

Requirement: All applicable Critical/High closure plus operational readiness. Measurement / pass threshold: Zero open Critical/High in enabled paths; audited scope/keys; resource limits; restore and incident drills pass. Responsible role: Security lead + operator lead. Evidence artifact: closure manifests, load/fault tests, alerts, backup recovery proof.

Failure response: stop promotion, retain evidence, contain the affected path, repair and independently re-run the failing acceptance cases. Waiver policy: no waiver for fund safety, consensus, key

management or state integrity; non-safety deviations require a signed scope reduction with owner and expiry, never a falsified pass. Current status: BLOCKED / NOT ESTABLISHED.

Incentivized testnet

Requirement: Public-testnet gates plus abuse-resistant reward accounting. Measurement / pass threshold: Every reward/slash reconciles; replay/Sybil abuse limits proven; no unresolved economic safety defect. Responsible role: Economic/security lead. Evidence artifact: ledger reconciliation, abuse test corpus and published reward policy.

Failure response: stop promotion, retain evidence, contain the affected path, repair and independently re-run the failing acceptance cases. Waiver policy: no waiver for fund safety, consensus, key management or state integrity; non-safety deviations require a signed scope reduction with owner and expiry, never a falsified pass. Current status: BLOCKED / NOT ESTABLISHED.

Release candidate

Requirement: Exact artifact reproducibility and independent review. Measurement / pass threshold: Two independent builders reproduce runtime WASM; clean source; all tests/gates pass; no stale evidence. Responsible role: Release manager + independent reviewer. Evidence artifact: signed source/build/feature/spec manifest and reviewer retest.

Failure response: stop promotion, retain evidence, contain the affected path, repair and independently re-run the failing acceptance cases. Waiver policy: no waiver for fund safety, consensus, key management or state integrity; non-safety deviations require a signed scope reduction with owner and expiry, never a falsified pass. Current status: BLOCKED / NOT ESTABLISHED.

Mainnet

Requirement: All previous gates, funded-route proof, custody and governance readiness. Measurement / pass threshold: Zero unresolved nonwaivable findings; genesis/keys/upgrade/restore approved; capacity SLO proven on final release. Responsible role: Accountable protocol/security/operator signers. Evidence artifact: signed launch decision covering exact binary/WASM/genesis hashes.

Failure response: stop promotion, retain evidence, contain the affected path, repair and independently re-run the failing acceptance cases. Waiver policy: no waiver for fund safety, consensus, key management or state integrity; non-safety deviations require a signed scope reduction with owner and expiry, never a falsified pass. Current status: BLOCKED / NOT ESTABLISHED.

Final truth statement

Demonstrably working today: source inventory/metadata extraction; four RPC middleware algorithm tests; Python intent parsing and selected typechecking/simulation tests. The adversarial router harness demonstrably reveals invalid-proof acceptance, and the Makefile experiment demonstrably reveals swallowed failure. These are bounded observations, not operating-network proof.

What only appears complete: a gateway dependency graph without a serving binary; proof names without authentic evidence; EVM-shaped RPC without canonical signed pool submission; synthetic mini-EVM state and SDK encoders; validator metadata penalties described as economic slashing; readiness gates satisfied by stale artifacts or masked statuses. Missing evidence includes finalized multi-node execution, authenticated enabled external settlement, full build/test success, contract runtime tests, restore/upgrade drills, custody validation and sustained network performance.

Before public testnet: close nonwaivable defects in enabled scope and execute the private-network gates. Before mainnet: add independent retest, reproducible signed artifacts, genesis/custody ceremony, operational and economic assurance, authenticated external routes if enabled, measured capacity and accountable approval. This audit changes no protocol code and grants no launch authorization.

64 SCOPED CAPABILITIES

Appendix A Detailed Feature Acceptance Cards

Each card corresponds exactly to a CSV/scorecard row. The completion figure is evidence-criteria coverage. Effort and dependency fields are planning estimates; commands identify where acceptance tests belong and do not imply that all proposed tests already exist.

FT01 Aura block production

IMPLEMENTED BUT UNVERIFIED | 40% | Consensus | P1

Evidence / entrypoint: node/src/service.rs:1059

Missing work / acceptance: Prove real multi-node authoring and partition recovery.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test -p x3-chain-node --lib
```

FT02 GRANDPA finality and fork choice

IMPLEMENTED BUT UNVERIFIED | 40% | Consensus | P1

Evidence / entrypoint: node/src/service.rs:601; node/src/service.rs:1136

Missing work / acceptance: Run finality, equivocation and restart tests.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT03 Validator rotation

PARTIAL | 20% | Consensus | P1

Evidence / entrypoint: pallets/x3-consensus/src/lib.rs:412

Missing work / acceptance: Fix activation-delay/session mismatch (H05).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT04 Validator bonded stake and slashing

PARTIAL | 20% | Consensus | P1

Evidence / entrypoint: pallets/x3-consensus/src/lib.rs:263

Missing work / acceptance: Authenticate offences and debit real stake; prove session removal (H04).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT05 Flash finality

PARTIAL | 0% | Consensus | P1

Evidence / entrypoint: crates/flash-finality/src/lib.rs:389; node/src/service.rs:656

Missing work / acceptance: Authority membership and a working replacement finalizer are required (H06).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT06 Genesis construction and live-seed validation

IMPLEMENTED BUT UNVERIFIED | 40% | Consensus | P1

Evidence / entrypoint: node/src/chain_spec.rs:760

Missing work / acceptance: Generate and independently validate production spec with actual operator public identities.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT07 Signed native transaction path

IMPLEMENTED BUT UNVERIFIED | 40% | Transactions | P1

Evidence / entrypoint: runtime/src/lib.rs:850; node/src/rpc.rs:1050

Missing work / acceptance: Execute wallet-to-pool-to-finalized-receipt transfer on two nodes.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT08 Nonce, genesis, era replay checks

IMPLEMENTED BUT UNVERIFIED | 40% | Transactions | P1

Evidence / entrypoint: runtime/src/lib.rs:850

Missing work / acceptance: Exercise wrong-chain, stale era, duplicate and concurrent nonces.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT09 Transaction pool limits and ordering

PARTIAL | 20% | Transactions | P1

Evidence / entrypoint: node/src/service.rs:589; node/src/service.rs:371

Missing work / acceptance: Preserve operator limits and measure spam/eviction behavior.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT10 Fee charging and refunds

IMPLEMENTED BUT UNVERIFIED | 40% | Transactions | P1

Evidence / entrypoint: runtime/src/lib.rs:1091

Missing work / acceptance: Reconcile charged fee, treasury credit, burn and failed-call refunds.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT11 Ethereum raw transaction RPC

PARTIAL | 20% | Transactions | P1

Evidence / entrypoint: node/src/rpc_frontier.rs:457

Missing work / acceptance: Replace custom query execution with signed pool submission (H08).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT12 Canonical FRAME storage

IMPLEMENTED BUT UNVERIFIED | 40% | State | P1

Evidence / entrypoint: node/src/service.rs:567; runtime/src/lib.rs:834

Missing work / acceptance: Crash/restart, state-root reconstruction and corrupted database drills.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT13 Storage upgrade migrations

BLOCKED | 40% | State | P1

Evidence / entrypoint: runtime/src/lib.rs:826

Missing work / acceptance: Run ordered old-schema upgrade and reverse/recovery proofs; full build blocked.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT14 Supply conservation ledger

PARTIAL | 20% | State | P1

Evidence / entrypoint: pallets/x3-supply-ledger/src/lib.rs:158

Missing work / acceptance: Bound finalization work and run cross-domain conservation properties (H16).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT15 Snapshot backup and restoration

PARTIAL | 20% | State | P1

Evidence / entrypoint: scripts/snapshot-restore.sh:130

Missing work / acceptance: Authenticate snapshot and validate restored finalized state (M04).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT16 Atomic rollback state provenance

PARTIAL | 20% | State | P1

Evidence / entrypoint: pallets/x3-atomic-kernel/src/lib.rs:1030

Missing work / acceptance: Reject unauthenticated diffs and demonstrate complete rollback (C03).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure;

relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT17 Signature primitives and key custody

IMPLEMENTED BUT UNVERIFIED | 40% | Crypto | P1

Evidence / entrypoint: node/src/service.rs:323; runtime/src/lib.rs:141

Missing work / acceptance: Prove hardware/custody signing and key rotation without dev fallback.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT18 Dependency advisory handling

PARTIAL | 20% | Crypto | P1

Evidence / entrypoint: .cargo/audit.toml:1; evidence/advisories-unfiltered.json

Missing work / acceptance: Triage all 53 advisory/package matches against shipped binaries (H15).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT19 P2P networking and sync

IMPLEMENTED BUT UNVERIFIED | 40% | Networking | P1

Evidence / entrypoint: node/src/service.rs:881

Missing work / acceptance: Four-node discovery, catch-up, reconnection and resource-exhaustion tests.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT20 Bootstrap helper configuration

DISCONNECTED | 20% | Networking | P1

Evidence / entrypoint: node/src/network.rs:9

Missing work / acceptance: Prove helper configuration is consumed by service; service uses `sc_network` configuration.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT21 PoH import verification

PARTIAL | 0% | Networking | P1

Evidence / entrypoint: node/src/service.rs:943

Missing work / acceptance: Prove all peer import paths validate consistent PoH rules before opting in.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT22 RPC limiter algorithm (narrow scope)

VERIFIED | 100% | Networking | P1

Evidence / entrypoint: node/src/rpc_middleware.rs:395; evidence/rpc-harness.log

Missing work / acceptance: Retain unit evidence; whole-node abuse resistance remains separate.

Tests: Observed passing for scoped behavior. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 1, 1, 1]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --manifest-path audit-harness/rpc/Cargo.toml
```

FT23 WASM mini-EVM persistent execution

PARTIAL | 20% | VM | P1

Evidence / entrypoint: crates/evm-integration/src/mini_evm.rs:102

Missing work / acceptance: Use actual accounts/state/caller/gas persistence (H07).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT24 SVM execution and account context

PARTIAL | 20% | VM | P1

Evidence / entrypoint: pallets/x3-kernel/src/wasm_adapters.rs:86

Missing work / acceptance: Pass real account input, slot, blockhash and signer context; prove CPI and persistence.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT25 X3 VM runtime

IMPLEMENTED BUT UNVERIFIED | 40% | VM | P1

Evidence / entrypoint: pallets/x3-kernel/src/wasm_adapters.rs:182

Missing work / acceptance: Prove runtime adapter semantics and differential native/WASM determinism.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT26 X3 Python parser and typechecker

VERIFIED | 100% | VM | P1

Evidence / entrypoint: x3-lang/cli.py:1; evidence/dsl-tests.log; evidence/dsl-cli.log

Missing work / acceptance: Parsing/typing only: does not prove trade execution.

Tests: Observed passing for scoped behavior. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 1, 1, 1]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
/usr/bin/python3 -m pytest x3-lang/tests/test_parser.py x3-lang/tests/test_typechecker.py
```

FT27 X3 emitter and execution pipeline

PARTIAL | 20% | VM | P1

Evidence / entrypoint: x3-lang/emitter/x3.py:60

Missing work / acceptance: Fix four failed emitter tests with authentic proof fixtures (M05).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT28 Rust compiler tracks

IMPLEMENTED BUT UNVERIFIED | 20% | VM | P1

Evidence / entrypoint: crates/x3-compiler/Cargo.toml:1; x3-lang/compiler/src/lib.rs:1

Missing work / acceptance: Reconcile separate workspaces and verify bytecode parity against actual VM.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT29 Internal cross-VM representation router

IMPLEMENTED BUT UNVERIFIED | 40% | Cross-chain | P1

Evidence / entrypoint: pallets/x3-cross-vm-router/src/lib.rs:1

Missing work / acceptance: Run six-route conservation/rollback matrix through full runtime.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT30 Atomic bundle orchestration / Atomic Lock

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: pallets/x3-atomic-kernel/src/lib.rs:839

Missing work / acceptance: Close C02/C03 before treating a PoAE hash as proof.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT31 Settlement timeout and refund engine

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: pallets/x3-settlement-engine/src/lib.rs:1350

Missing work / acceptance: Authenticate receipt/finality inputs; prove timely refunds under load.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT32 External header finality oracle

PLACEHOLDER | 20% | Cross-chain | P1

Evidence / entrypoint: pallets/cross-chain-validator/src/lib.rs:332

Missing work / acceptance: Replace structural header and byte-count quorum checks (C01).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT33 Production EVM receipt proof route

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-verification-router/src/evm_receipt.rs:737

Missing work / acceptance: Trusted receipts root and canonical inclusion/receipt-value binding (H02).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT34 Solana finalized proof route

PLACEHOLDER | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-verification-router/src/lib.rs:355

Missing work / acceptance: Replace nonempty check with authenticated finality and event proofs (H01).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT35 Validator quorum proof route

PLACEHOLDER | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-verification-router/src/lib.rs:323

Missing work / acceptance: Verify distinct authorized signatures over exact envelope (H01).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT36 Bitcoin SPV route

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-verification-router/src/lib.rs:545

Missing work / acceptance: Verify best-work chain, correct confirmation height and locked output (H03).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT37 Bitcoin vault approvals

PLACEHOLDER | 0% | Cross-chain | P1

Evidence / entrypoint: crates/x3-bitcoin-vault/src/lib.rs:324

Missing work / acceptance: Replace call-count progression and unverified signatures (H03).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT38 Relay binary delivery

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-relayer/src/main.rs:1442

Missing work / acceptance: Persist retries/cursors and prove finalized proof submission (H09/H10).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT39 Relayer typed submission library

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-relayer/src/submitter.rs:568

Missing work / acceptance: Use correct metadata/custody/signed-extension serialization (H09).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT40 EVM HTLC contract

IMPLEMENTED BUT UNVERIFIED | 20% | Cross-chain | P1

Evidence / entrypoint: X3-contracts/evm/contracts/AtlasHTLC.sol:99

Missing work / acceptance: Run claim/refund/expiry/reentrancy and nonstandard-token accounting tests.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
forge test --root X3-contracts/evm
```

FT41 SVM HTLC contract

IMPLEMENTED BUT UNVERIFIED | 20% | Cross-chain | P1

Evidence / entrypoint: X3-contracts/svm/programs/x3_htlc/src/lib.rs:1

Missing work / acceptance: Compile Anchor program and prove local validator claim/refund parity.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
anchor test
```

FT42 External EVM proof verifier contract

PARTIAL | 0% | Cross-chain | P1

Evidence / entrypoint: X3-contracts/evm/contracts/EvmReceiptVerifier.sol:205

Missing work / acceptance: Implement actual signature verification without mode bypass (H17).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT43 Solver marketplace and intent routing

IMPLEMENTED BUT UNVERIFIED | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-intent/Cargo.toml:1; pallets/atomic-trade-engine/src/lib.rs:1

Missing work / acceptance: Trace signed intent to competing solver selection and bonded settlement.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 0, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT44 Validator attestation / proof ledger

PARTIAL | 20% | Cross-chain | P1

Evidence / entrypoint: crates/x3-validator-attestation/src/lib.rs:1;
pallets/x3-atomic-kernel/src/lib.rs:1416

Missing work / acceptance: Bind signer set, execution evidence and finality to authenticated runtime state.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT45 REST/GraphQL gateway

DISCONNECTED | 0% | Operations | P1

Evidence / entrypoint: crates/x3-gateway/src/main.rs:1

Missing work / acceptance: Wire executable configuration, DB/router/HTTP server (H11).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT46 SQL indexer and migrations

DISCONNECTED | 20% | Operations | P1

Evidence / entrypoint: crates/x3-gateway/src/db.rs:1

Missing work / acceptance: Activate indexer and migrate disposable DB end to end; gateway modules not compiled.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT47 Metrics / chain health monitoring

PARTIAL | 20% | Observability | P1

Evidence / entrypoint: node/src/metrics.rs:1; crates/x3-gateway/src/rest.rs:248

Missing work / acceptance: Measure dependency health, finality lag and sink delivery (M03).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT48 Security and accounting event consumers

DISCONNECTED | 0% | Observability | P1

Evidence / entrypoint: runtime/src/lib.rs:11

Missing work / acceptance: Implement durable consumer wiring; logs explicitly report dropped events.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT49 DEX, Forge and LP locker

IMPLEMENTED BUT UNVERIFIED | 40% | Governance | P1

Evidence / entrypoint: pallets/x3-dex/src/lib.rs:1; pallets/x3-token-factory/src/lib.rs:1; pallets/x3-lp-locker/src/lib.rs:1

Missing work / acceptance: Verify swaps/mints/locks/fees and EconomicHalt under full runtime.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT50 Sentinel and economic halt

PARTIAL | 20% | Governance | P1

Evidence / entrypoint: pallets/x3-sentinel/src/lib.rs:1; pallets/x3-invariants/src/lib.rs:618

Missing work / acceptance: Prove halt scope covers privileged/unsigned paths and refunds remain available.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build

closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT51 Council / treasury / upgrades

IMPLEMENTED BUT UNVERIFIED | 40% | Governance | P1

Evidence / entrypoint: runtime/src/lib.rs:543; runtime/src/lib.rs:826

Missing work / acceptance: Prove threshold, timelock, session/treasury separation and upgrade recovery.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT52 Wallet / biometric / recovery pallet

PARTIAL | 20% | Crypto | P1

Evidence / entrypoint: pallets/x3-wallet-pallet/src/lib.rs:263

Missing work / acceptance: Independently audit recovery authorization, privacy and biometric replay resistance.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT53 TypeScript SDK encoding

PLACEHOLDER | 20% | Transactions | P1

Evidence / entrypoint: packages/ts-sdk/src/evm.ts:154; packages/ts-sdk/src/svm.ts:230

Missing work / acceptance: Canonical selectors, addresses and account-message compilation (H12).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Safety critical. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT54 Desktop / Tauri OS network integration

PARTIAL | 20% | Operations | P1

Evidence / entrypoint: apps/x3-desktop/src/blockchain/ChainManager.ts:59;
apps/tauri-os/package.json:1

Missing work / acceptance: Repair default network lookup; render and verify real signing/receipt flows (M06).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build

closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT55 GPU / swarm orchestration

PARTIAL | 0% | Operations | P1

Evidence / entrypoint: node/src/service.rs:1244; crates/x3-swarm-core/Cargo.toml:1

Missing work / acceptance: Prove actual workload execution, required-feature startup failure and restart behavior.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT56 CI test quality and coverage

PARTIAL | 20% | Tests | P1

Evidence / entrypoint: Makefile:19; node/src/service.rs:2664

Missing work / acceptance: Stop swallowed failures and require real multi-node tests (H13/M02).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT57 Workspace build / release reproducibility

BLOCKED | 0% | Deployment | P1

Evidence / entrypoint: evidence/check-complete.log; rust-toolchain.toml:3

Missing work / acceptance: Resolve duplicate-core WASM failure and reproduce artifacts (H18).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT58 Fresh-machine bootstrap

BLOCKED | 0% | Deployment | P1

Evidence / entrypoint: scripts/fresh_machine_check.sh:5; Makefile:61

Missing work / acceptance: Pin toolchain/dependencies and correct package/target names (L01).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```


FT59 Deployment isolation and validator services

IMPLEMENTED BUT UNVERIFIED | 40% | Operations | P1

Evidence / entrypoint: packaging/systemd/x3-validator.service:1

Missing work / acceptance: Run least-privilege validator installation and backup/restore rehearsal.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [1, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT60 Sustained finalized TPS evidence

MISSING | 0% | Performance | P3

Evidence / entrypoint: crates/x3-bench/src/runner.rs:251

Missing work / acceptance: Run controlled network workload; compiler throughput is not chain TPS (M07).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT61 Mainnet / ProofGate enforcement

PARTIAL | 20% | Proof gates | P1

Evidence / entrypoint: scripts/mainnet_release_gate.py:84; Makefile:19

Missing work / acceptance: Bind every gate to source/features/artifacts and propagate failures (H13/H14).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 1, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT62 Documentation accuracy / scoreboard

PARTIAL | 0% | Documentation | P1

Evidence / entrypoint: CURRENT_MAINNET_STATUS.md:3; FEATURE_REGISTRY.toml:18

Missing work / acceptance: Derive operational status from fresh executable proofs (M01).

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT63 Public testnet recovery drill evidence

MISSING | 0% | Operations | P1

Evidence / entrypoint: node/src/service.rs:2664; scripts/snapshot-restore.sh:130

Missing work / acceptance: Produce four-validator partition/rejoin and restored finalized-state evidence.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

FT64 Independent launch approval evidence

MISSING | 0% | Proof gates | P1

Evidence / entrypoint: LAUNCH_SCOPE.md:1

Missing work / acceptance: Obtain external security review, closure retest and operator-signed release decision.

Tests: No passing production-path evidence in this audit. Criteria [implemented, wired, tested, executed, reproducible] = [0, 0, 0, 0, 0]. Security: Operational integrity. Dependencies: H18 build closure; relevant finding dependencies. Effort: L (ordinal estimate; not a delivery promise).

```
cargo test --workspace
```

[COMMANDS](#) / [EXIT CODES](#) / [ARTIFACTS](#)

Appendix B Evidence Ledger and Repository Inventories

All verification attempts below are retained, including failed first attempts and interrupted/timed-out compilation. Initial 180-second attempts timed out; continued checks established E0152 failures. Test counts are only reported when an actual test runner emitted them. Read-only exploratory shell commands are also in the conversation tool transcript; the durable ledger below focuses on proof-producing checks and declared limitations.

Table 15 Verification command ledger

ID / exit	Command	Evidence
tools / 0	rustc --version; cargo --version; node --version; npm --version; pnpm --version; python --version; python3 --version; /usr/bin/python3 --version; cargo audit --version; forge --version; anchor --version; protoc --version; clang --version	evidence/tools.log
metadata / 0	cargo metadata --offline --locked --no-deps --format-version 1	evidence/metadata.log
check / 124	cargo check --workspace	evidence/check.log
test / 124	cargo test --workspace	evidence/test.log
clippy / 124	cargo clippy --workspace --all-targets -- -D warnings	evidence/clippy.log
pnpm-test / 127	pnpm test	evidence/pnpm-test.log
pnpm-build / 127	pnpm build	evidence/pnpm-build.log
npm-test / 127	npm test	evidence/npm-test.log
pytest / 127	python -m pytest	evidence/pytest.log
fmt / 1	cargo fmt --all -- --check	evidence/fmt.log
rpc-unit / 1	rustc --edition=2021 --test node/src/rpc_middleware.rs -o /t mp/x3-audit-20260905/rpc-tests && /tmp/x3-audit-20260905/rpc- tests --test-threads=1	evidence/rpc-unit.log
dsl-tests / 1	/usr/bin/python3 -m pytest -q x3-lang/tests/test_parser.py x3-l ang/tests/test_typechecker.py x3-lang/tests/test_emitter.py x3-lang/tests/test_simulator.py	evidence/dsl-tests.log
dsl-cli / 0	/usr/bin/python3 x3-lang/cli.py x 3-lang/examples/arb_solana_eth. x3	evidence/dsl-cli.log
fresh / 1	bash scripts/fresh_machine_check.sh	evidence/fresh.log
consistency / 0	bash scripts/check-readiness-con sistency.sh	evidence/consistency.log
npm-build / 127	npm run build	evidence/npm-build.log
audit-help / 0	cargo audit --help	evidence/audit-help.log
python-collection / 2	/usr/bin/python3 -m pytest --collect-only -q tests/test_migrations.py	evidence/python-collection.log

ID / exit	Command	Evidence
check-complete / 101	cargo check --workspace	evidence/check-complete.log
test-complete / 101	cargo test --workspace	evidence/test-complete.log
clippy-complete / 101	cargo clippy --workspace --all-targets -- -D warnings	evidence/clippy-complete.log
release-complete / 101	cargo build --locked --release -p x3-chain-node --features mainnet-rc1	evidence/release-complete.log
testnet-complete / 101	cargo check --locked -p x3-chain-node --features testnet	evidence/testnet-complete.log
contracts / 127	forge test --root X3-contracts/evm --offline	evidence/contracts.log
anchor / 127	anchor test --skip-deploy --skip-local-validator	evidence/anchor.log
rpc-harness / 0	CARGO_NET_OFFLINE=true CAR GO_TARGET_DIR=/tmp/x3-audit- 20260905/rpc-target cargo test --manifest-path /tmp/x3-audit-20 260905/rpc-harness/Cargo.toml	evidence/rpc-harness.log
proof-harness / 101	CARGO_NET_OFFLINE=true CAR GO_TARGET_DIR=/tmp/x3-audit- 20260905/proof-target cargo test --manifest-path /tmp/x3-audi t-20260905/proof-harness/Cargo. toml	evidence/proof-harness.log
gate-wallet / 0	CARGO_NET_OFFLINE=true CAR GO_TARGET_DIR=/tmp/x3-audit- 20260905/gate-target make test-x3-wallet	evidence/gate-wallet.log
missing-target / 2	make -n test-node-build	Conversation tool output / interrupted attempt
advisories-configured / 0	cargo audit --no-fetch --stale --no-yanked --json	evidence/advisories.json
advisories-unsuppressed / 1	cargo audit --no-fetch --stale --no-yanked --file /tmp/x3-audit-2 0260905/advisory-unfiltered/Car go.lock --json (cwd /tmp)	evidence/advisories-unfiltered.json
advisories-copy-error / 1	cp Cargo.lock /tmp/x3-audit-202 60905/advisory-unfiltered/Cargo. lock (cwd /tmp)	Conversation tool output / interrupted attempt
initial-release-interrupted / 130	cargo build --locked --release -p x3-chain-node --features mainnet-rc1	Conversation tool output / interrupted attempt

ID / exit	Command	Evidence
fake-scan / 2	<pre>grep -RIn TODO\ FIXME\ stub\ mock\ fake\ placeholder\ dummy\ unimplemented!\ todo!\ panic!("\n ot implemented . --exclude-dir=.git --exclude-dir=target --exclude-dir=node_modules</pre>	evidence/fake-scan-locations.jsonl

Environment: rustc 1.90.0 (1159e78c4), cargo 1.90.0, Node v22.23.2, npm 10.9.8, Python 3.14.7 on python3 PATH and Python 3.10.12 at /usr/bin/python3; ReportLab 3.6.8; protobuf 36.0; clang 14.0.0. pnpm, python alias, forge and anchor were unavailable on PATH. System Python had pytest; Alembic was unavailable. Cargo ran offline against cached dependencies; advisory DB freshness was not established.

Table 16 Durable inventory map

Artifact	Purpose
evidence/component-map.json	145 workspace package targets, features and dependencies.
evidence/inventory.json	First-party path/manifests/scripts/category inventory; 375 manifests.
evidence/dependency-inventory.json	1,996 Cargo.lock package records; not a reachability map.
evidence/configuration-inventory.txt	Configuration paths; secret values intentionally excluded.
evidence/migrations-inventory.txt	Migration candidates across runtime, SQL and Python.
evidence/api-inventory.txt	RPC/gateway source inventory; route membership needs endpoint context.
evidence/contracts-inventory.txt	EVM/SVM contract/program source inventory.
evidence/workflows-inventory.txt	CI workflow inventory; no hosted branch-protection assertion.
evidence/tests-inventory.txt	Test-path inventory, not coverage or passing status.
evidence/source-hashes.json	6,909 inspected text-file hashes, excluding credential/build/vendor scopes.
evidence/working-tree-before.txt	Pre-existing dirty source state. Commit alone does not reproduce it.

Workspace package directory

Table 17 Workspace package targets

Package	Manifest / targets
pallet-x3-kernel	pallets/x3-kernel/Cargo.toml pallet_x3_kernel (rlib)
pallet-x3-invariants	pallets/x3-invariants/Cargo.toml pallet_x3_invariants (rlib)
x3-security-events	crates/x3-security-events/Cargo.toml x3_security_events (rlib)
x3-cross-vm-bridge	crates/cross-vm-bridge/Cargo.toml x3_cross_vm_bridge (rlib), integration (test)
x3-evm-integration	crates/evm-integration/Cargo.toml x3_evm_integration (lib), erc20_integration (test), integration (test)
x3-packet-schema	crates/x3-packet-schema/Cargo.toml x3_packet_schema (lib)
x3-svm-integration	crates/svm-integration/Cargo.toml x3_svm_integration (lib), counter_integration (test)
svm_counter	crates/svm-counter/Cargo.toml svm_counter (cdylib/rlib)
x3-x3-integration	crates/x3-integration/Cargo.toml x3_x3_integration (lib), compiler_bridge (test)
x3-backend	crates/x3-backend/Cargo.toml x3_backend (lib)
x3-ast	crates/x3-ast/Cargo.toml x3_ast (lib)
x3-common	crates/x3-common/Cargo.toml x3_common (lib)
x3-hir	crates/x3-hir/Cargo.toml x3_hir (lib)
x3-typeck	crates/x3-typeck/Cargo.toml x3_typeck (lib), golden (test)
x3-semantic	crates/x3-semantic/Cargo.toml x3_semantic (lib), golden (test)
x3-parser	crates/x3-parser/Cargo.toml x3_parser (lib), golden (test)
x3-lexer	crates/x3-lexer/Cargo.toml x3_lexer (lib)
x3-mir	crates/x3-mir/Cargo.toml x3_mir (lib)
x3-compiler	crates/x3-compiler/Cargo.toml x3_compiler (lib), determinism (test), e2e_test (test), integration_test (test), schema_validator (test)
x3-opt	crates/x3-opt/Cargo.toml x3_opt (lib), loop_pack_integration_bench (test), optimizer_yolo_smoke (test)
x3-vm	crates/x3-vm/Cargo.toml x3_vm (lib), gpu_integration (test)
x3-verifier	crates/x3-verifier/Cargo.toml x3_verifier (lib), integration (test)
invariant-macros	crates/invariant-macros/Cargo.toml invariant_macros (proc-macro), registry_check (test), trybuild (test)
pallet-atomic-trade-engine	pallets/atomic-trade-engine/Cargo.toml pallet_atomic_trade_engine (lib)
pallet-governance	pallets/governance/Cargo.toml pallet_governance (rlib)
pallet-treasury	pallets/treasury/Cargo.toml pallet_treasury (rlib)

Package	Manifest / targets
pallet-agent-accounts	pallets/agent-accounts/Cargo.toml pallet_agent_accounts (rlib)
pallet-agent-memory	pallets/agent-memory/Cargo.toml pallet_agent_memory (rlib)
pallet-svm	pallets/svm-runtime/Cargo.toml pallet_svm (lib)
pallet-evolution-core	pallets/evolution-core/Cargo.toml pallet_evolution_core (lib)
pallet-x3-verifier	pallets/x3-verifier/Cargo.toml pallet_x3_verifier (lib)
pallet-x3-domain-registry	pallets/x3-domain-registry/Cargo.toml pallet_x3_domain_registry (lib)
pallet-x3-settlement-engine	pallets/x3-settlement-engine/Cargo.toml pallet_x3_settlement_engine (lib), property_tests (test)
pallet-x3-atomic-kernel	pallets/x3-atomic-kernel/Cargo.toml pallet_x3_atomic_kernel (lib), e2e_settlement (test), loom_concurrency (test), miri_tests (test), proptest_tests (test)
x3-asset-kernel-types	crates/x3-asset-kernel-types/Cargo.toml x3_asset_kernel_types (rlib)
x3-crosschain-intent	crates/x3-crosschain-intent/Cargo.toml x3_crosschain_intent (lib), compiler (test), end_to_end (test)
pallet-meme-overlord	pallets/meme-overlord/Cargo.toml pallet_meme_overlord (lib)
pallet-swarm	pallets/swarm/Cargo.toml pallet_swarm (rlib)
pallet-x3-wallet	pallets/x3-wallet-pallet/Cargo.toml pallet_x3_wallet (lib)
x3-wallet	crates/x3-wallet/Cargo.toml x3_wallet (lib)
x3-gateway	crates/x3-gateway/Cargo.toml x3-gateway (bin), loom_mempool_concurrency (test), shuttle_validator_async (test)
x3-bridge	crates/x3-bridge/Cargo.toml x3_bridge (lib)
x3-orchestra-control-plane	crates/x3-orchestra-control-plane/Cargo.toml x3_orchestra_control_plane (lib), x3-orchestra-control-plane (bin)
x3-relayer	crates/x3-relayer/Cargo.toml x3_relayer (lib), x3-relayer (bin)
x3-finality-oracle	crates/x3-finality-oracle/Cargo.toml x3_finality_oracle (lib)
x3-gateway-risk-engine	crates/x3-gateway-risk-engine/Cargo.toml x3_gateway_risk_engine (lib)
x3-proof-dispute	crates/x3-proof-dispute/Cargo.toml x3_proof_dispute (lib)
x3-validator-attestation	crates/x3-validator-attestation/Cargo.toml x3_validator_attestation (lib)
x3-verification-router	crates/x3-verification-router/Cargo.toml x3_verification_router (lib)
x3-bitcoin-vault	crates/x3-bitcoin-vault/Cargo.toml x3_bitcoin_vault (lib)
x3-rpc	crates/x3-rpc/Cargo.toml x3_rpc (lib)
pallet-x3-crosschain-gateway	pallets/x3-crosschain-gateway/Cargo.toml pallet_x3_crosschain_gateway (lib)
tps-tracker	crates/tps-tracker/Cargo.toml tps_tracker (lib), tps-tracker (bin)

Package	Manifest / targets
x3-chain-runtime	runtime/Cargo.toml x3_chain_runtime (rlib), fraud_proofs_proptest (test), fraud_proofs_witness_v1 (test), build-script-build (custom-build)
pallet-cross-chain-validator	pallets/cross-chain-validator/Cargo.toml pallet_cross_chain_validator (lib)
pallet-depin-marketplace	pallets/depin-marketplace/Cargo.toml pallet_depin_marketplace (rlib)
pallet-northern-swarm	pallets/northern-swarm/Cargo.toml pallet_northern_swarm (lib)
pallet-private-execution	pallets/private-execution/Cargo.toml pallet_private_execution (rlib)
pallet-x3-account-registry	pallets/x3-account-registry/Cargo.toml pallet_x3_account_registry (rlib)
pallet-x3-agent-law	pallets/x3-agent-law/Cargo.toml pallet_x3_agent_law (lib)
pallet-x3-agent-registry	pallets/pallet-x3-agent-registry/Cargo.toml pallet_x3_agent_registry (rlib)
x3-accounting-events	crates/x3-accounting-events/Cargo.toml x3_accounting_events (rlib)
pallet-x3-asset-registry	pallets/x3-asset-registry/Cargo.toml pallet_x3_asset_registry (rlib)
pallet-x3-auction	pallets/x3-auction/Cargo.toml pallet_x3_auction (rlib)
pallet-x3-automation	pallets/x3-automation/Cargo.toml pallet_x3_automation (lib)
x3-automation	crates/x3-automation/Cargo.toml x3_automation (lib)
pallet-x3-coin	pallets/x3-coin/Cargo.toml pallet_x3_coin (lib)
pallet-x3-compute-market	pallets/x3-compute-market/Cargo.toml pallet_x3_compute_market (rlib)
pallet-x3-consensus	pallets/x3-consensus/Cargo.toml pallet_x3_consensus (lib)
pallet-x3-cross-vm-router	pallets/x3-cross-vm-router/Cargo.toml pallet_x3_cross_vm_router (rlib)
x3-ixl	crates/x3-ixl/Cargo.toml x3_ixl (lib), properties (test)
x3-packet-standard	crates/x3-packet-standard/Cargo.toml x3_packet_standard (lib), properties (test)
x3-liquidity-core	crates/x3-liquidity-core/Cargo.toml x3_liquidity_core (rlib)
x3-dex	crates/x3-dex/Cargo.toml x3_dex (lib)
pallet-x3-supply-ledger	pallets/x3-supply-ledger/Cargo.toml pallet_x3_supply_ledger (rlib)
pallet-x3-custody	pallets/x3-custody/Cargo.toml pallet_x3_custody (rlib)
pallet-x3-da	pallets/x3-da/Cargo.toml pallet_x3_da (lib)
pallet-x3-dapp-hub	pallets/x3-dapp-hub/Cargo.toml pallet_x3_dapp_hub (rlib)
x3-revenue-sharing	crates/x3-revenue-sharing/Cargo.toml x3_revenue_sharing (lib)
pallet-x3-dex	pallets/x3-dex/Cargo.toml pallet_x3_dex (lib)
pallet-x3-flashloan	pallets/x3-flashloan/Cargo.toml pallet_x3_flashloan (rlib)
pallet-x3-inventory	pallets/x3-inventory/Cargo.toml pallet_x3_inventory (lib)

Package	Manifest / targets
pallet-x3-jury-anchor	pallets/x3-jury-anchor/Cargo.toml pallet_x3_jury_anchor (lib)
pallet-x3-launchpad	pallets/x3-launchpad/Cargo.toml pallet_x3_launchpad (rlib)
pallet-x3-lp-locker	pallets/x3-lp-locker/Cargo.toml pallet_x3_lp_locker (rlib)
pallet-x3-oracle	pallets/x3-oracle/Cargo.toml pallet_x3_oracle (lib)
pallet-x3-partner	pallets/x3-partner/Cargo.toml pallet_x3_partner (lib)
pallet-x3-proof-carrying-agent	pallets/pallet-x3-proof-carrying-agent/Cargo.toml pallet_x3_proof_carrying_agent (rlib)
pallet-x3-rebalance	pallets/x3-rebalance/Cargo.toml pallet_x3_rebalance (lib)
pallet-x3-reconciliation	pallets/x3-reconciliation/Cargo.toml pallet_x3_reconciliation (rlib)
pallet-x3-reservation	pallets/x3-reservation/Cargo.toml pallet_x3_reservation (lib)
pallet-x3-sentinel	pallets/x3-sentinel/Cargo.toml pallet_x3_sentinel (rlib)
pallet-x3-sequencer	pallets/x3-sequencer/Cargo.toml pallet_x3_sequencer (lib)
pallet-x3-slash	pallets/x3-slash/Cargo.toml pallet_x3_slash (lib)
x3-proof	crates/x3-proof/Cargo.toml x3_proof (lib), x3-proof (bin)
x3-slash	crates/x3-slash/Cargo.toml x3_slash (lib)
pallet-x3-solvency	pallets/x3-solvency/Cargo.toml pallet_x3_solvency (lib)
pallet-x3-token-factory	pallets/x3-token-factory/Cargo.toml pallet_x3_token_factory (rlib)
pallet-x3-treasury-policy	pallets/x3-treasury-policy/Cargo.toml pallet_x3_treasury_policy (lib)
pallet-x3-vrf	pallets/x3-vrf/Cargo.toml pallet_x3_vrf (lib)
x3-vrf	crates/x3-vrf/Cargo.toml x3_vrf (lib)
pallet-x3-wrapped	pallets/x3-wrapped/Cargo.toml pallet_x3_wrapped (rlib)
quantum-crypto	crates/quantum-crypto/Cargo.toml quantum_crypto (lib)
x3-metrics-tracker	crates/x3-metrics-tracker/Cargo.toml x3_metrics_tracker (rlib)
x3-staking-analytics	crates/x3-staking-analytics/Cargo.toml x3_staking_analytics (lib)
x3-chain-node	node/Cargo.toml x3_chain_node (lib), x3-chain-node (bin), flash_finality_network (test), node_requirements (test), rpc_dex_latency (bench)
contention-predictor	crates/contention-predictor/Cargo.toml contention_predictor (lib), inference_latency (bench)
cross-chain-gpu-validator	crates/cross-chain-gpu-validator/Cargo.toml cross_chain_gpu_validator (lib), cross-chain-gpu-validator (bin), integration_tests (test), gpu_kernels (bench)

Package	Manifest / targets
x3-gpu-validator-swarm	crates/x3-gpu-validator-swarm/Cargo.toml x3_gpu_validator_swarm (lib), wallet_sync (bin), x3-cpu-validator (bin), x3-swarm-bench (bin), x3-swarm-orchestrator (bin), x3-validator (bin), chaos_stress_test (test), metrics_sliding_window_integration (test), stress_harness (test), stress_with_real_time_metrics (test), test_x3_validator (test), tps_sliding_window_test (test), accel_sha256 (bench), e2e_tps (bench), swarm_tps (bench)
x3-accel	crates/x3-accel/Cargo.toml x3_accel (lib)
x3-accel-wgpu	crates/x3-accel-wgpu/Cargo.toml x3_accel_wgpu (lib)
x3-flash-finality	crates/flash-finality/Cargo.toml x3_flash_finality (lib)
parallel-proposer	crates/parallel-proposer/Cargo.toml parallel_proposer (lib), authoring_overhead (bench)
substrate-prometheus-endpoint	patches/substrate-prometheus-endpoint/Cargo.toml substrate_prometheus_endpoint (lib)
x3-poh-generator	crates/poh-generator/Cargo.toml x3_poh_generator (lib)
x3-atomic-trade	crates/x3-atomic-trade/Cargo.toml x3_atomic_trade (lib)
x3-bridge-adapters	crates/x3-bridge-adapters/Cargo.toml x3_bridge_adapters (lib)
analytics-service	apps/analytics/analytics-service/Cargo.toml analytics-service (bin)
pallet-fraud-proofs	pallets/fraud-proofs/Cargo.toml pallet_fraud_proofs (lib)
e2e_tests	tests/e2e/Cargo.toml e2e_tests (lib), cross_vm_real_chain_test (test), gateway_integration_test (test), internal_mainnet_happy_path (test), live_internal_mainnet_e2e (test), mainnet_rc1 (test)
x3-fees	crates/x3-fees/Cargo.toml x3_fees (lib), prop_fee_invariants (test)
x3-universal-contracts	crates/x3-universal-contracts/Cargo.toml x3_universal_contracts (lib)
x3-intent	crates/x3-intent/Cargo.toml x3_intent (lib)
launchops	tools/launchops/Cargo.toml launchops (bin)
proof-forge	proof-forge/Cargo.toml proof_forge (lib), x3-proof (bin)
x3-proving-harness	proving/Cargo.toml x3-prove (bin)
x3-atomic-swap	crates/x3-atomic-swap/Cargo.toml x3_atomic_swap (lib), gen_wallet (bin), atlas_htlc_deploy_test (test), atomic_swap_chaos (test), atomic_swap_integration (test), rpc_live_test (test)
x3-readiness	crates/x3-readiness/Cargo.toml x3_readiness (lib), x3-readiness (bin)
x3-svm	crates/x3-svm/Cargo.toml x3_svm (lib)
northern-swarm	crates/northern-swarm/Cargo.toml northern_swarm (lib), northern-swarm (bin)
x3-circuit-breaker	crates/x3-circuit-breaker/Cargo.toml x3_circuit_breaker (lib)
x3-external-route-registry	crates/x3-external-route-registry/Cargo.toml x3_external_route_registry (lib)

Package	Manifest / targets
x3-gateway-indexer	crates/x3-gateway-indexer/Cargo.toml x3_gateway_indexer (lib)
x3-gateway-insurance	crates/x3-gateway-insurance/Cargo.toml x3_gateway_insurance (lib)
x3-proof-envelope	crates/x3-proof-envelope/Cargo.toml x3_proof_envelope (lib)
x3-crosschain-gateway	crates/x3-crosschain-gateway/Cargo.toml x3_crosschain_gateway (lib)
x3-court	crates/x3-court/Cargo.toml x3_court (lib)
x3-agent	crates/x3-agent/Cargo.toml x3_agent (lib)
x3-bridge-security-council	crates/x3-bridge-security-council/Cargo.toml x3_bridge_security_council (lib)
x3-genesis-builder	crates/x3-genesis-builder/Cargo.toml x3_genesis_builder (lib), x3-genesis-builder (bin)
x3-foundry-core	crates/x3-foundry-core/Cargo.toml x3_foundry_core (lib)
x3-foundry-auditor	crates/x3-foundry-auditor/Cargo.toml x3_foundry_auditor (lib)
x3-foundry-revenue	crates/x3-foundry-revenue/Cargo.toml x3_foundry_revenue (lib)
x3-foundry-indexer	crates/x3-foundry-indexer/Cargo.toml x3_foundry_indexer (lib)
x3-chain-health-daemon	crates/x3-chain-health-daemon/Cargo.toml x3-chain-health-daemon (bin)

CONTEXTUAL FINDINGS / RAW CANDIDATES

Appendix C Fake-Completeness and Unknowns

A text match is not a defect. Test fixtures, comments, UI input placeholders, vendored examples and historical documents must be distinguished from reachable production behavior. The first-party context scan records 15,967 candidate lines. The requested recursive grep recorded 54,017 matching locations but returned exit 2; its stderr was not retained, so exhaustive successful traversal cannot be claimed.

Table 18 Confirmed misleading or nonfunctional behavior

Area	Finding / interpretation
Cross-chain header proof	C01: root ignored and quorum derived from bytes.
Finality/execution proof	C02/C03: untrusted anchors and rollback diffs; public-field commitment is not execution proof.
Proof-router alternatives	H01: actual production-feature tests accepted arbitrary bytes.
Bitcoin vault state machine	H03: confirmation/approval progression by calls; helper SPV code does not cure that path.
Mini EVM	H07: synthetic accounts/state, empty changes; real interpreter does not imply real persistent VM.
Gateway	H11: executable logs then exits; source API handlers do not run.
SDK	H12: rolling hash, sliced address and zero account indices.
External EVM verifier	H17: production signature path blocks; test mode checks structure only.
Observability	M03: liveness 200 and logged dropped events, not durable dependency health.
Performance/readiness	M01/M07: path/mode/compile metrics do not prove network readiness.
Intent simulator	Explicit heuristic simulator; legitimate planning tool, not execution/finality proof.
Test-only mocks	Not automatically production defects; do not award operational credit for mocked tests.

Full candidate locations are shipped rather than falsely classifying all matches as confirmed findings. Manual contextual review prioritized runtime safety, bridge proof and accounting paths, startup wiring, SDK encoding and release gates. It did not manually review every line of all 11,620 inventoried paths or vendor source. Unreviewed code, unreachable historical copies, live deployments, credentials and external operator practices remain unknown. This limitation does not weaken the demonstrated NO-GO blockers, but prevents claiming a complete security certification.

No fresh network advisory fetch, independent cryptographic audit, mainnet balance observation, live database migration, fuzz campaign, full browser render of product UIs or multi-node runtime experiment completed. Those are explicit recovery tasks. A failed local build is environmental/source-integration evidence for this exact checkout, not proof that no alternate machine or existing binary has ever run X3.

TERMS / STATUS LABELS / EVIDENCE QUALITY

Appendix D Glossary and Reading Guide

Use status labels literally. VERIFIED applies only to the row scope and executed evidence. IMPLEMENTED BUT UNVERIFIED means meaningful code exists but its complete production behavior was not demonstrated. PARTIAL means a required part is missing or incorrect. PLACEHOLDER means structural/simulated/nonfunctional behavior stands in for the promised implementation. DISCONNECTED means code is not wired into the stated runtime. MISSING means no meaningful implementation/evidence of the scoped capability was found. BLOCKED means verification could not proceed, with its blocker recorded.

Table 19 Glossary

Term	Meaning in this report
Aura	Slot-based block-authoring consensus used by node service.
GRANDPA	Finality gadget wired as the default node finalizer.
Finality	A block commitment beyond mere head observation or local execution.
FRAME / pallet	Substrate runtime framework / runtime state-transition module.
Extrinsic / SignedExtra	Encoded runtime input / signed validation extensions including nonce, era, genesis and fees.
HTLC	Hash time-locked contract: claim by secret, refund after deadline under defined rules.
PoAE	Project proof-of-atomic-execution record; authenticity must be separately established.
SPV / MPT	Simplified payment verification / Merkle Patricia trie inclusion proof.
WASM / EVM / SVM	Runtime execution format / Ethereum virtual machine / Solana-style virtual machine.
RPO / RTO	Recovery point objective / recovery time objective; not measured here.
NO-GO	Do not promote the audited state to the named release scope.
Evidence score	Weighted coverage of explicit implementation/wiring/test/execution/reproduction criteria.

Evidence quality labels: Confirmed by execution means a command/test directly observed behavior; Confirmed by static inspection means source establishes the property; Inferred means a stated consequence or planning estimate; Claimed by documentation is a proposition being evaluated; Not verified/Blocked earns no runtime credit. File references refer to the audited working tree, whose selected content hashes are retained.

REPRODUCTION / CHECKSUMS / SOURCE STATE

Appendix E References, Regeneration and Integrity

Audited base commit: 6a24d8cf38f2522ddf9ae0b47011fd59a9984208. Branch: master. Audit date: 2026-09-05. Existing tracked and untracked edits are part of the reviewed state. No repository remote URL was returned; manifest repository URLs are claims and were not substituted for a verified origin.

Primary protocol references consulted: Ethereum JSON-RPC specification

<https://ethereum.org/developers/docs/apis/json-rpc/> ; Ethereum transaction model

<https://ethereum.org/developers/docs/transactions> ; Ethereum block/receiptsRoot model

<https://ethereum.org/developers/docs/blocks/> ; Polkadot runtime API documentation

<https://paritytech.github.io/polkadot-sdk/book/node/utility/runtime-api.html> . These support protocol terminology and the distinction between query, signed submission, transaction and receipt. Repository findings remain tied to exact local source. Cached RustSec matches require fresh official advisory verification and binary-specific reachability analysis before release.

Regenerate the booklet from this source package with `/usr/bin/python3 visuals.py`, `/usr/bin/python3 author_report.py`, then `/usr/bin/python3 render_report.py`. The source builder consumes `findings.json`, `scorecard.json`, `provenance.json` and evidence inventories; it never fetches data or executes deployments. See `README.md` for safe harness commands, source-hash checks and prerequisites. A clean checkout of the commit alone is insufficient because pre-existing changes were audited.

`manifest.json` lists each generated deliverable, purpose, UTC creation time, audited commit and SHA-256. It deliberately does not hash itself to avoid a circular checksum. `manifest.sha256` authenticates the manifest bytes; authenticity still requires a trusted distribution channel or signature by an accountable publisher. The final PDF hash cannot be embedded inside its own PDF without a circular dependency; use the external manifest.

Source and evidence files are durable project artifacts. Protocol implementation files were not edited. Audit completion means this evidence-based assessment and blueprint package was delivered; it does not mean the blockchain or every blocked experiment is complete.